

Sesión 2 - Secuenciación Masiva Aplicaciones

Isabel Cuesta

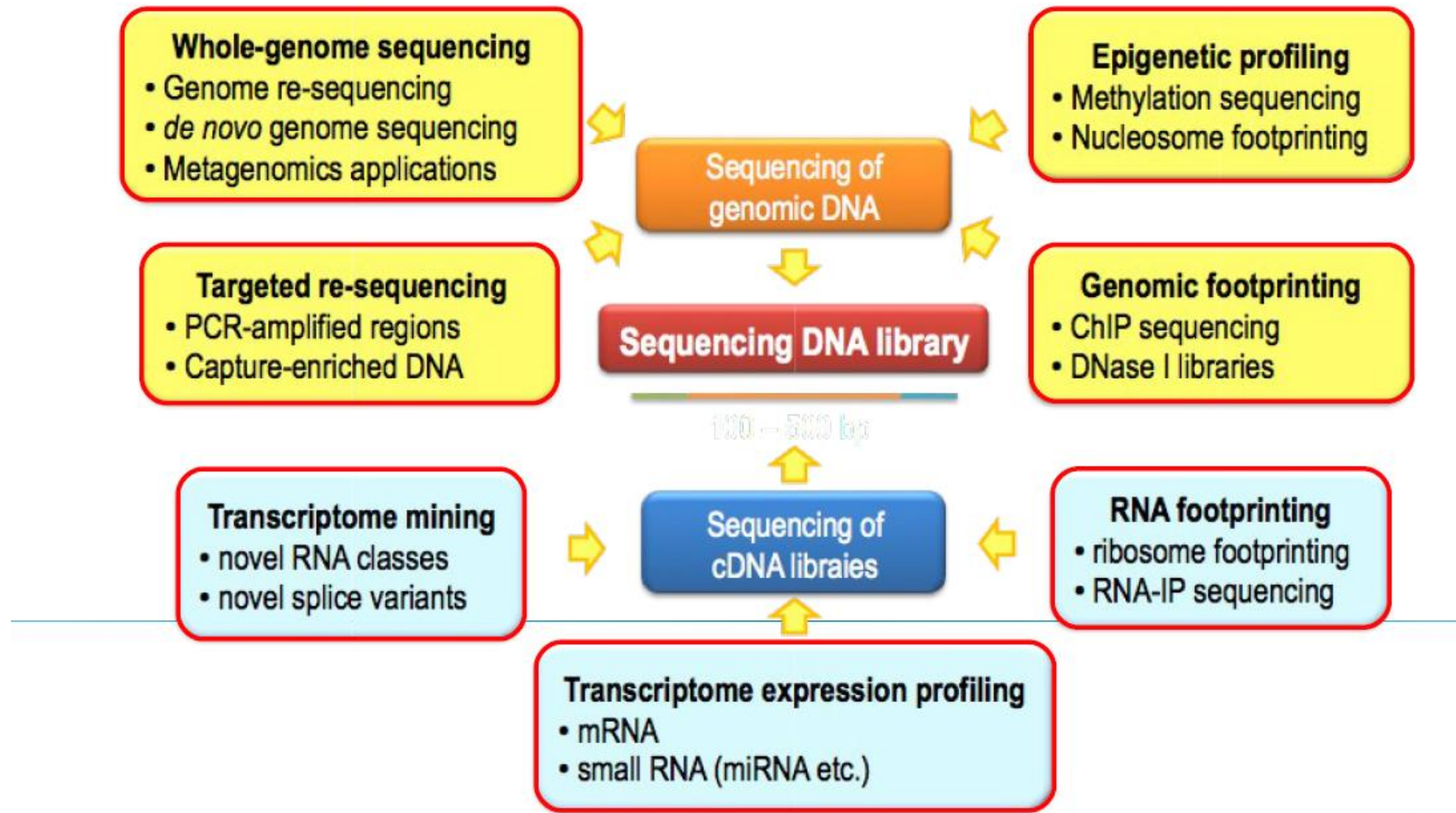
Unidad de Bioinformática (BU-ISCIII)
Unidades Centrales Científico Técnicas – SGSAFI-ISCIII

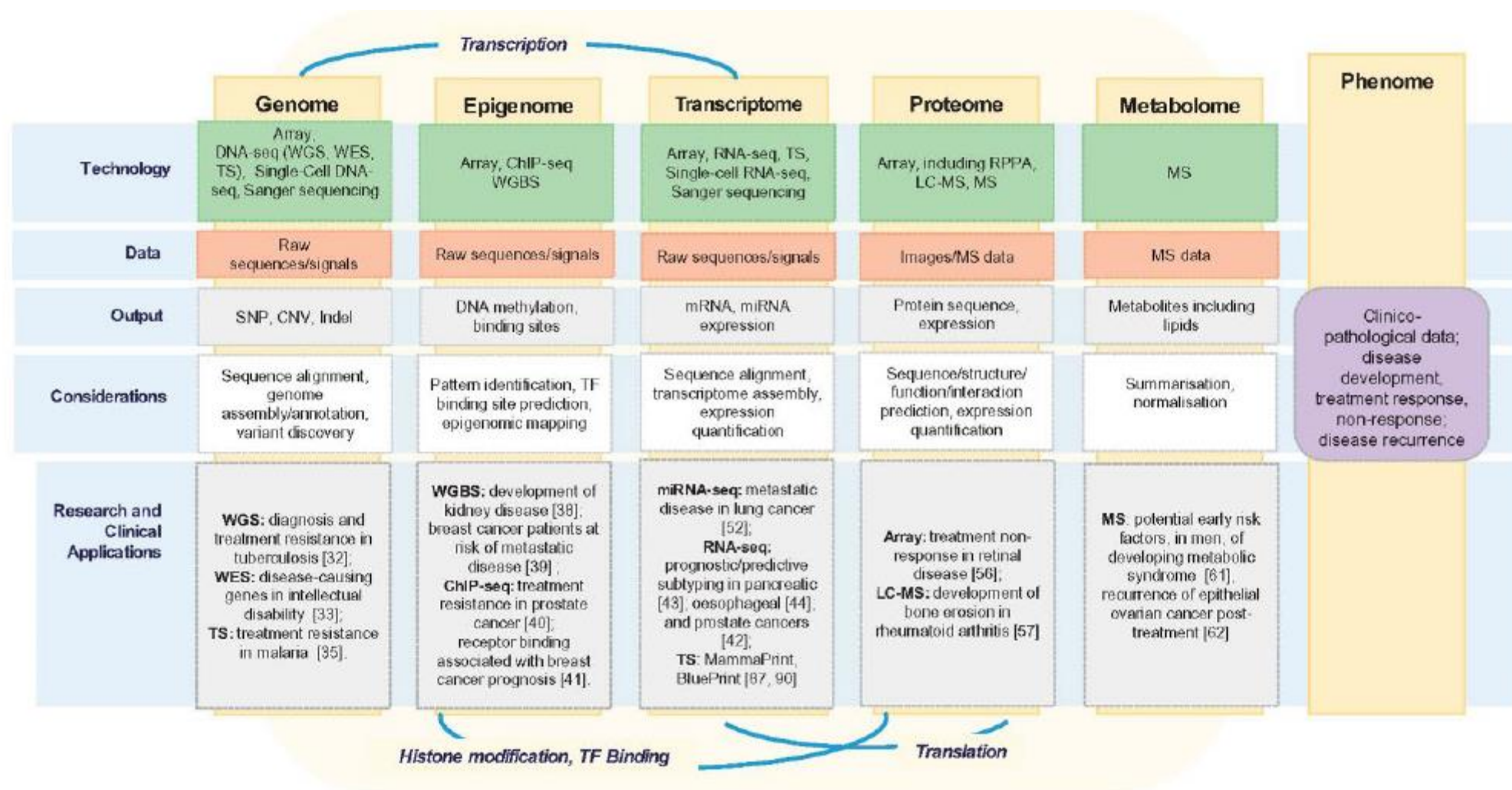
9 – 13 junio 2025, 12ª Edición
Programa Formación Continua, ISCIII

What has NGS changed?

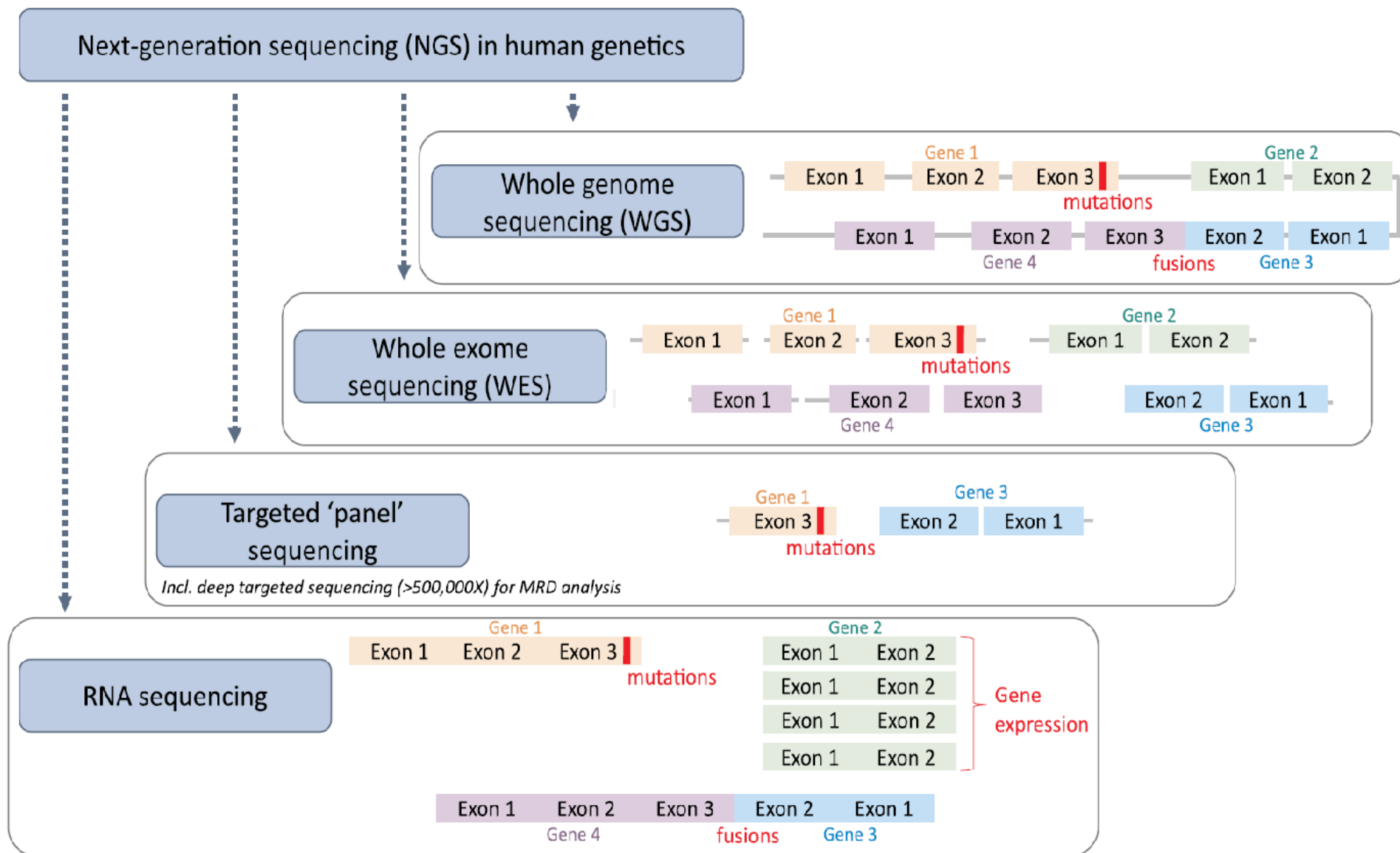
- ✓ **Functional genomics. Genome-Seq. Epigenetics**
- ✓ **Molecular diagnostics. Complex diseases**
- ✓ **Microbial Ecology. Metagenomics**
- ✓ **Molecular Ecology. Population Genetics**
- ✓ **Evolutionary Genomics**
- ✓ **DNA-Protein Interactions. ChIPSeq**
- ✓ **Pharmacogenomics**
- ✓ **Transcriptomics. RNAseq**
- ✓ **Systems Biology**

Aplicaciones de la secuenciación masiva

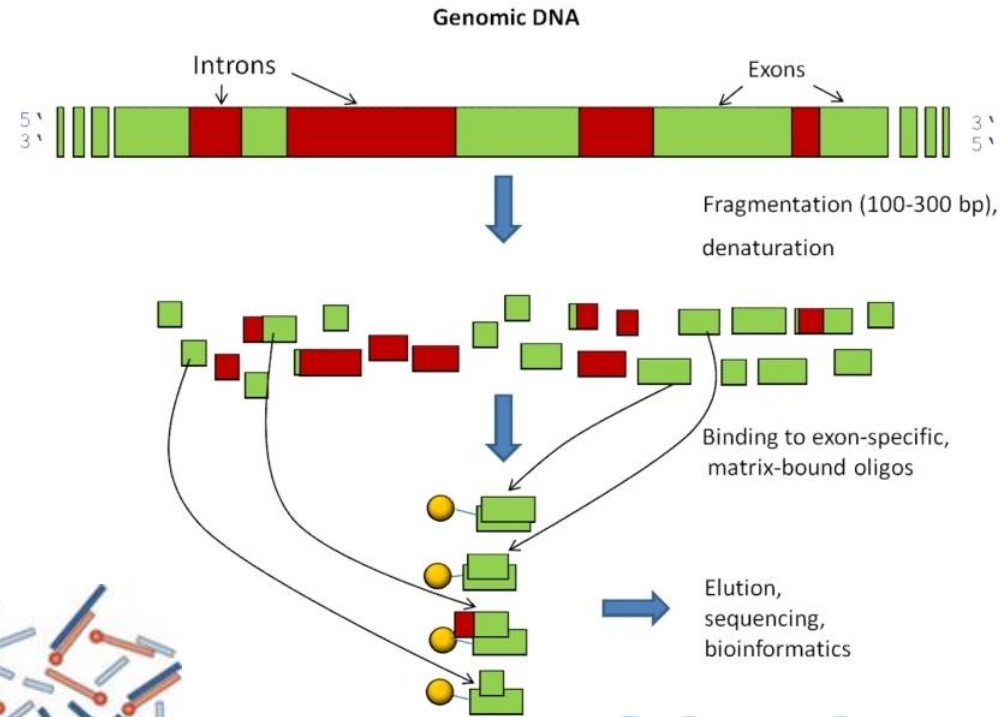
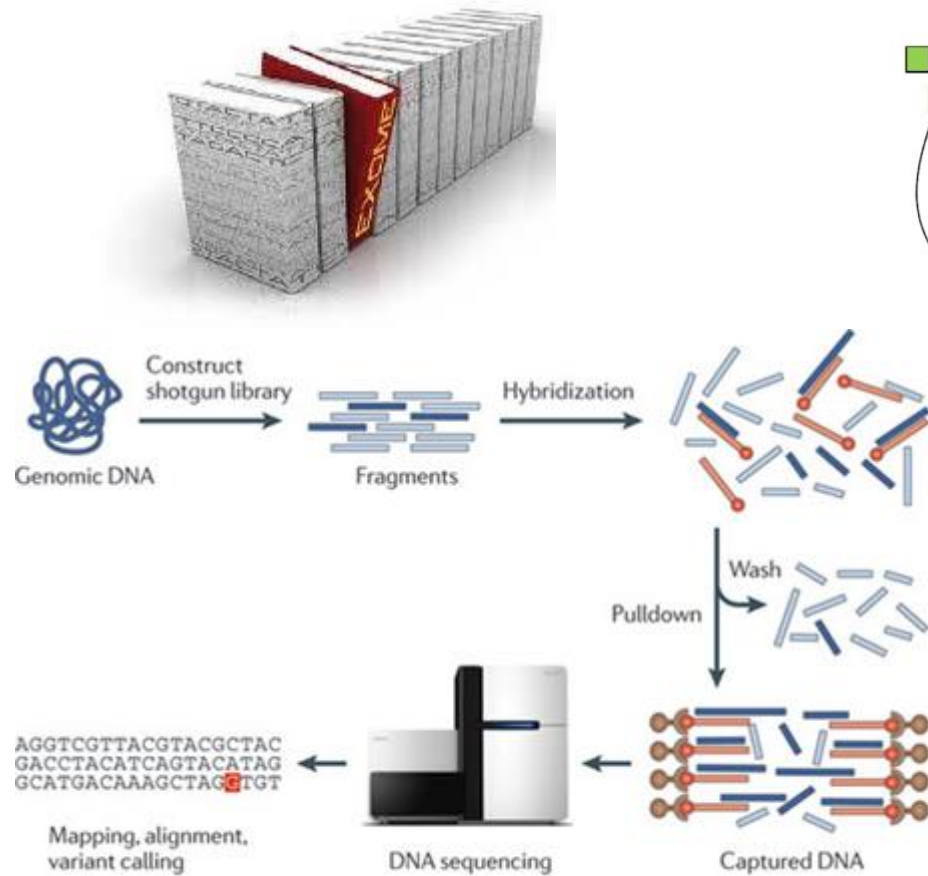




Human Genetics Applications



EXOME



30-50 Mb
Human Exome
\$500

Genoma, Exoma, Panel? desde un punto de vista clínico

PANEL

- Barato y rápido
- Util en enfermedades monogénicas
- Datos mas manejables, análisis y almacenamiento

EXOMA

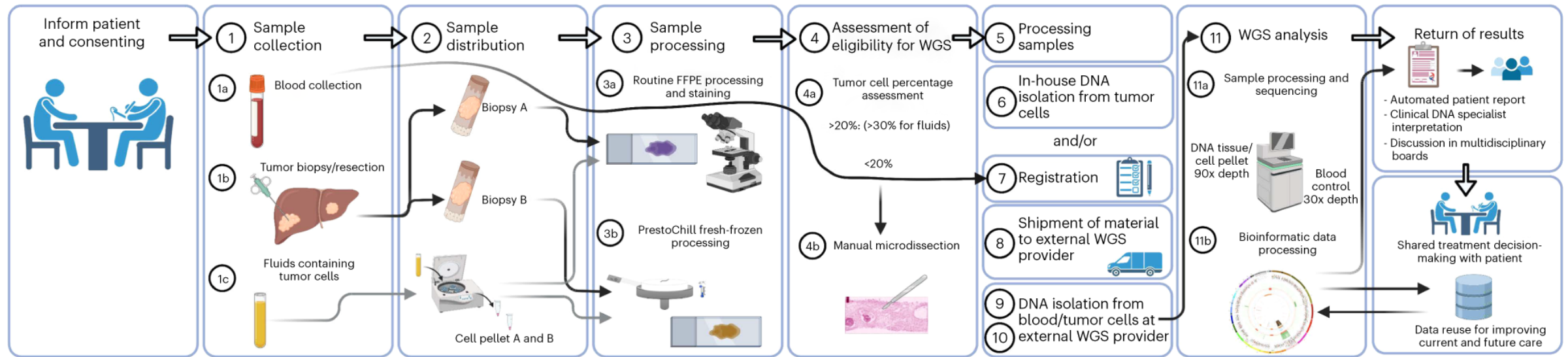
- Mas complejo y lento
- Necesario en enfermedades complejas
- Análisis mas complejo
- Mayor volumen de datos

GENOMA

- Maxima complejidad en secuenciación y coste
- Información de regiones no codificantes
- Análisis de variaciones estructurales
- Elevado volumen de datos

PRECISION ONCOLOGY

Overview of the procedures for performing WGS in routine tumor diagnostics



Samsom et al., Nature Protocols 2024, 19,700-726

PRECISION ONCOLOGY

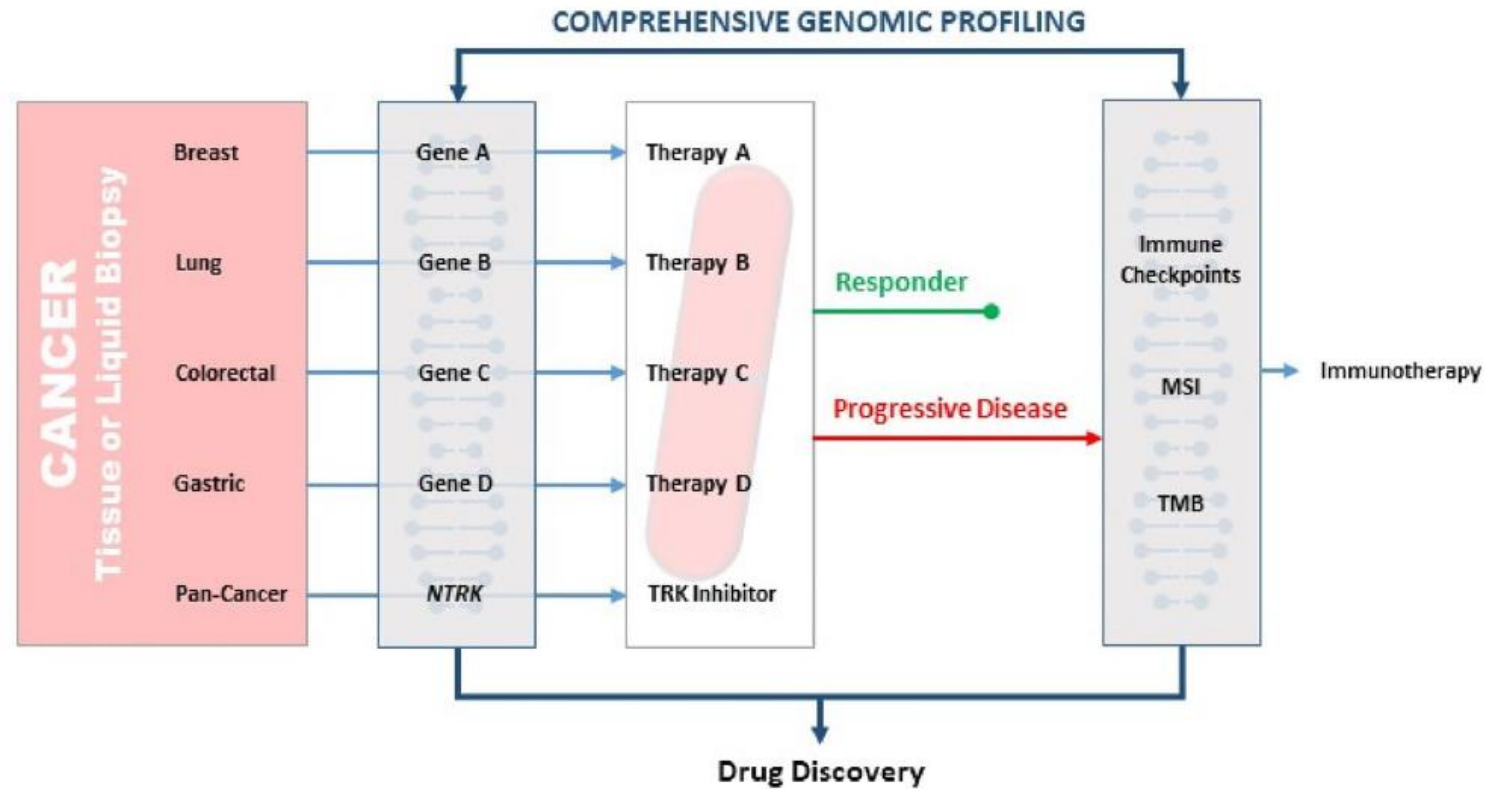


Fig. 1 Graphical overview of how Comprehensive Genomic Profiling leads to oncological advancements by precision medicine. *MSI* microsatellite instability, *TMB* tumor mutational burden

Pankiw et al., Medical Oncology 2024, 41:1

PRECISION ONCOLOGY

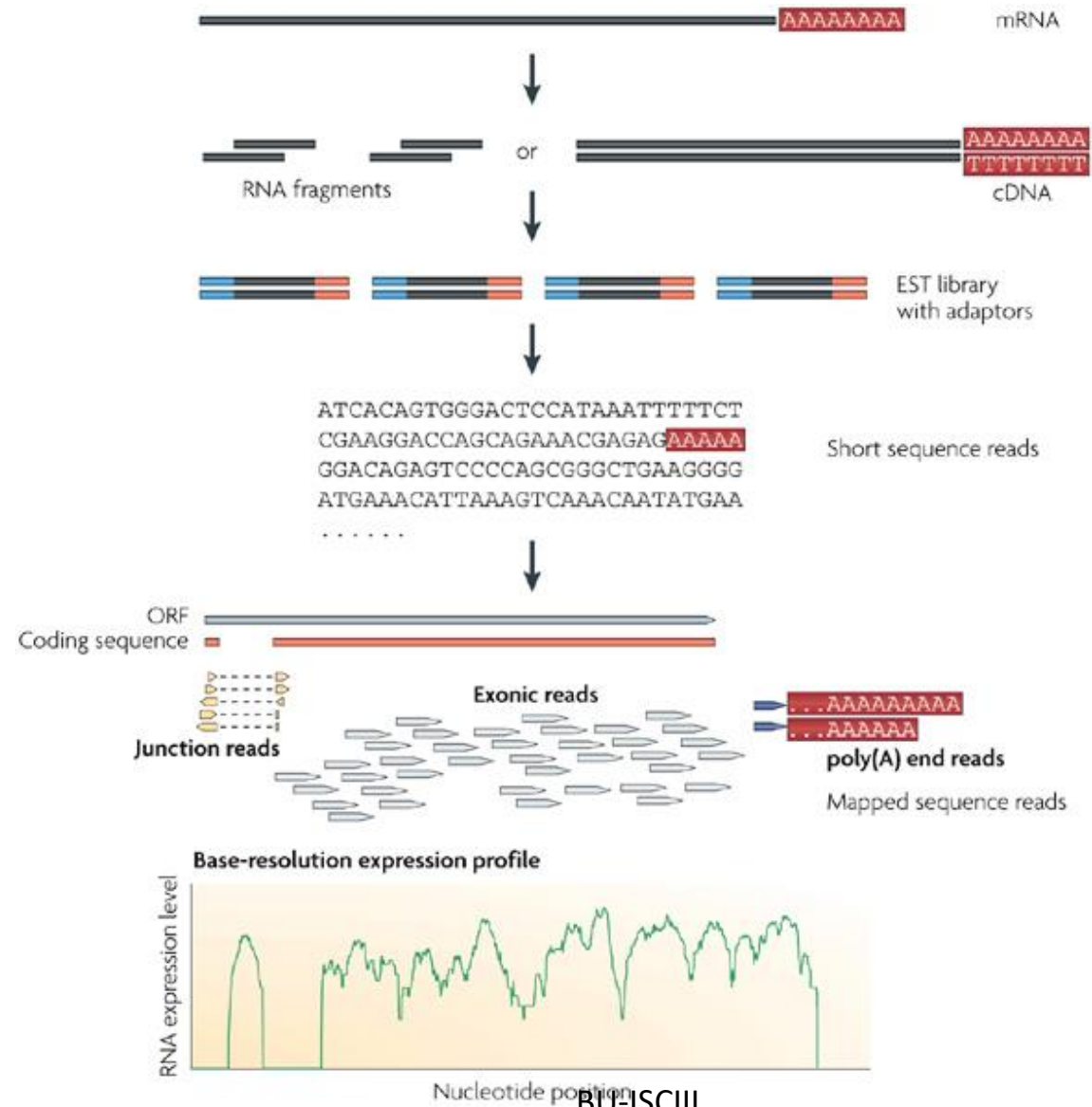
Benefits of comprehensive genomic profiling in oncology

Hallmark	Key aspects
Personalized Medicine	By providing a detailed genetic profile of a tumor, CGP supports the implementation of personalized medicine, where treatments are tailored to the unique characteristics of each patient’s cancer
Improved Treatment Outcomes	Patients who receive therapy based on CGP findings can often experience better outcomes, including higher response rates, longer progression-free survival, and, in some cases, improved overall survival
Cost-Effectiveness	Although initially more expensive than single-gene tests, CGP can be more cost-effective in the long term by identifying the most effective therapy from the outset, reducing the need for trial-and-error treatment approaches
Identification of Resistance Mechanisms	CGP can uncover genetic alterations that confer resistance to certain treatments, guiding clinicians in selecting alternative therapies or combination strategies to overcome resistance
Challenges and Considerations on CGP	Despite its advantages, CGP faces challenges, including the need for sufficient tumor tissue samples, the complexity of data interpretation, and the requirement for specialized bioinformatics support to analyze and understand the results. Additionally, there are ongoing discussions regarding the reimbursement and cost coverage of CGP tests by healthcare systems and insurance providers

CGP: comprehensive genomic profiling.

Zalis et al., Frontiers in Genetics, 2024, review

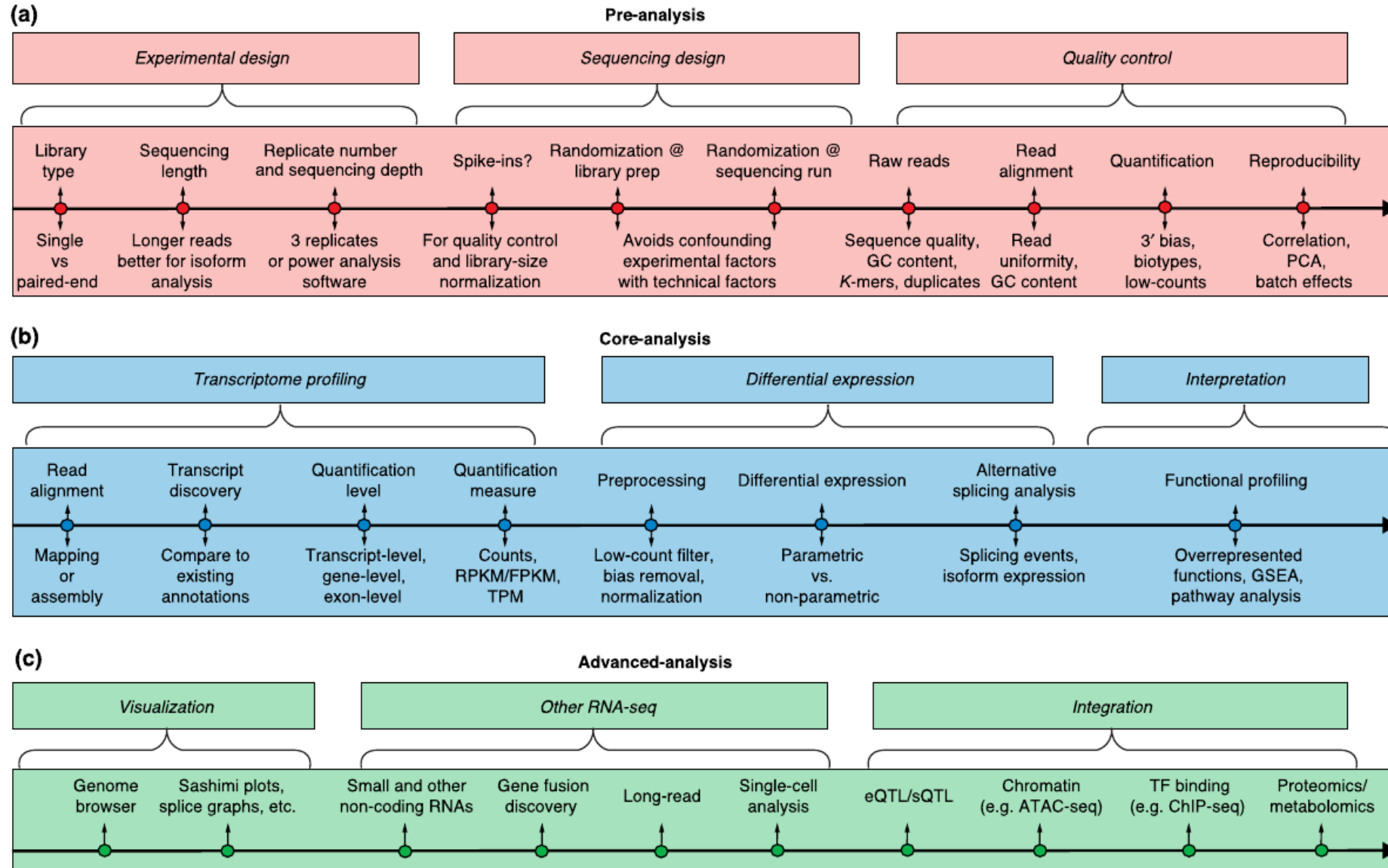
RNA seq



10/06/2025

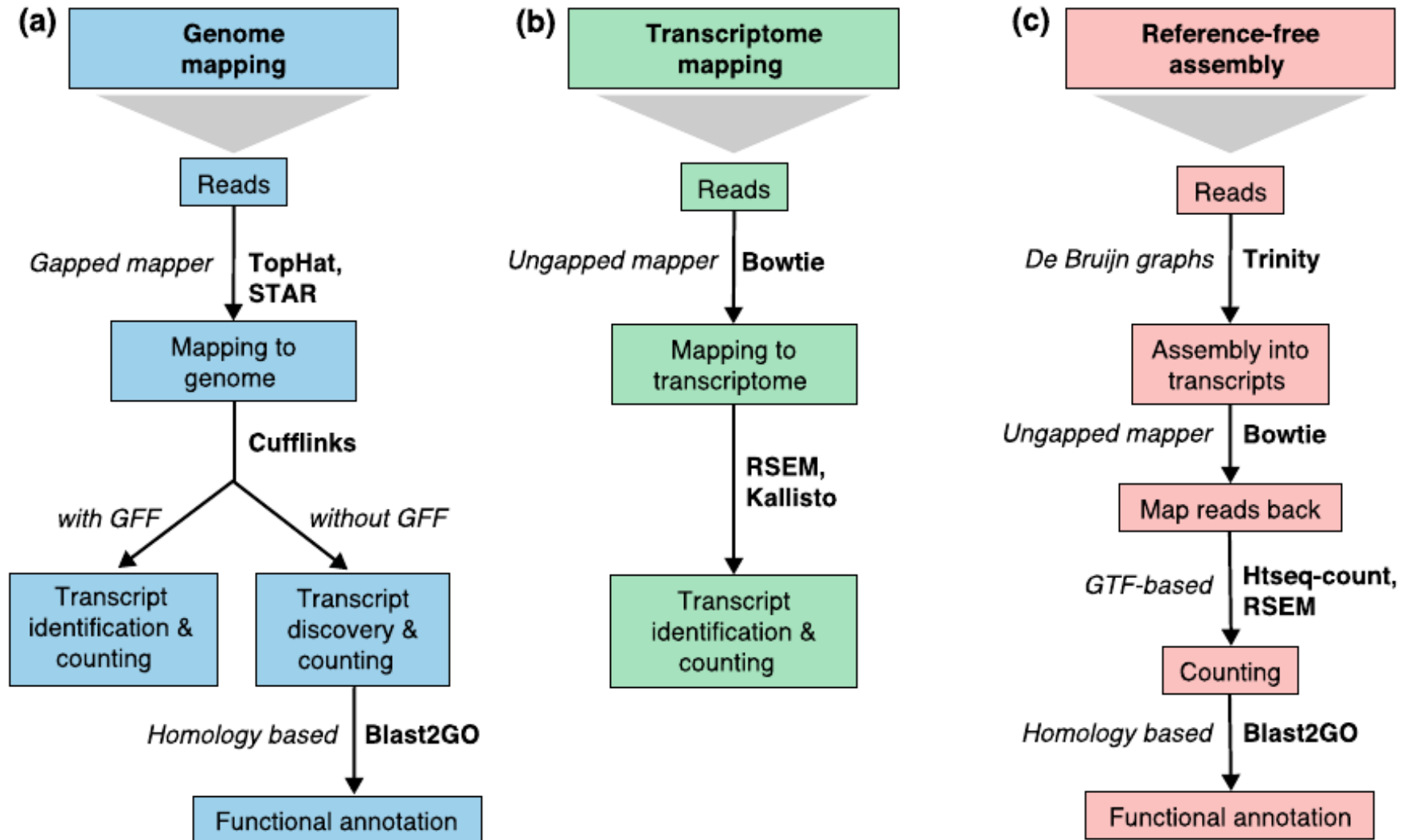
BU-ISCI

RNA seq



BU-ISCIII Conesa et al., Genome Biology (2016) 17:13

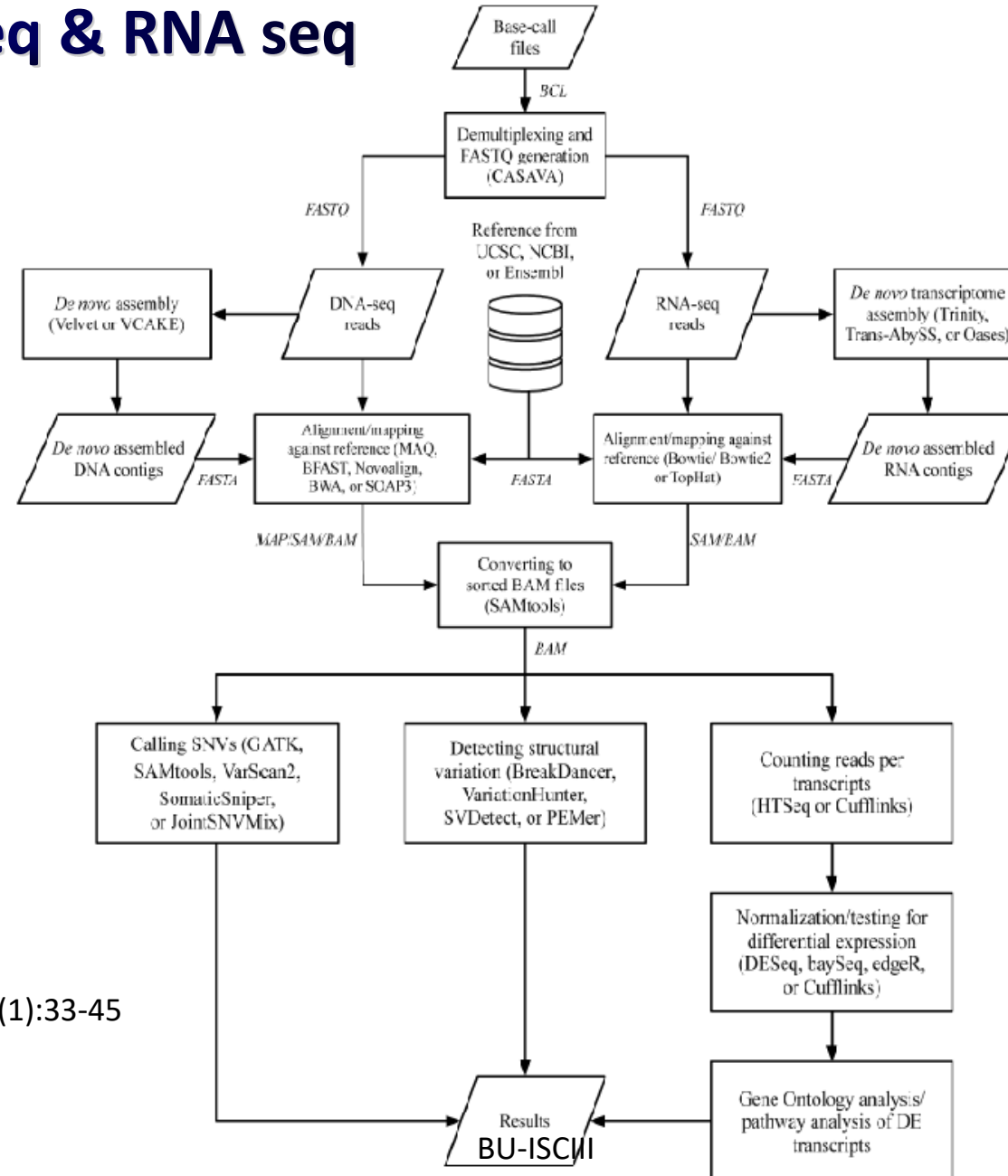
RNA seq: transcript identification strategies



RNA seq, ventajas

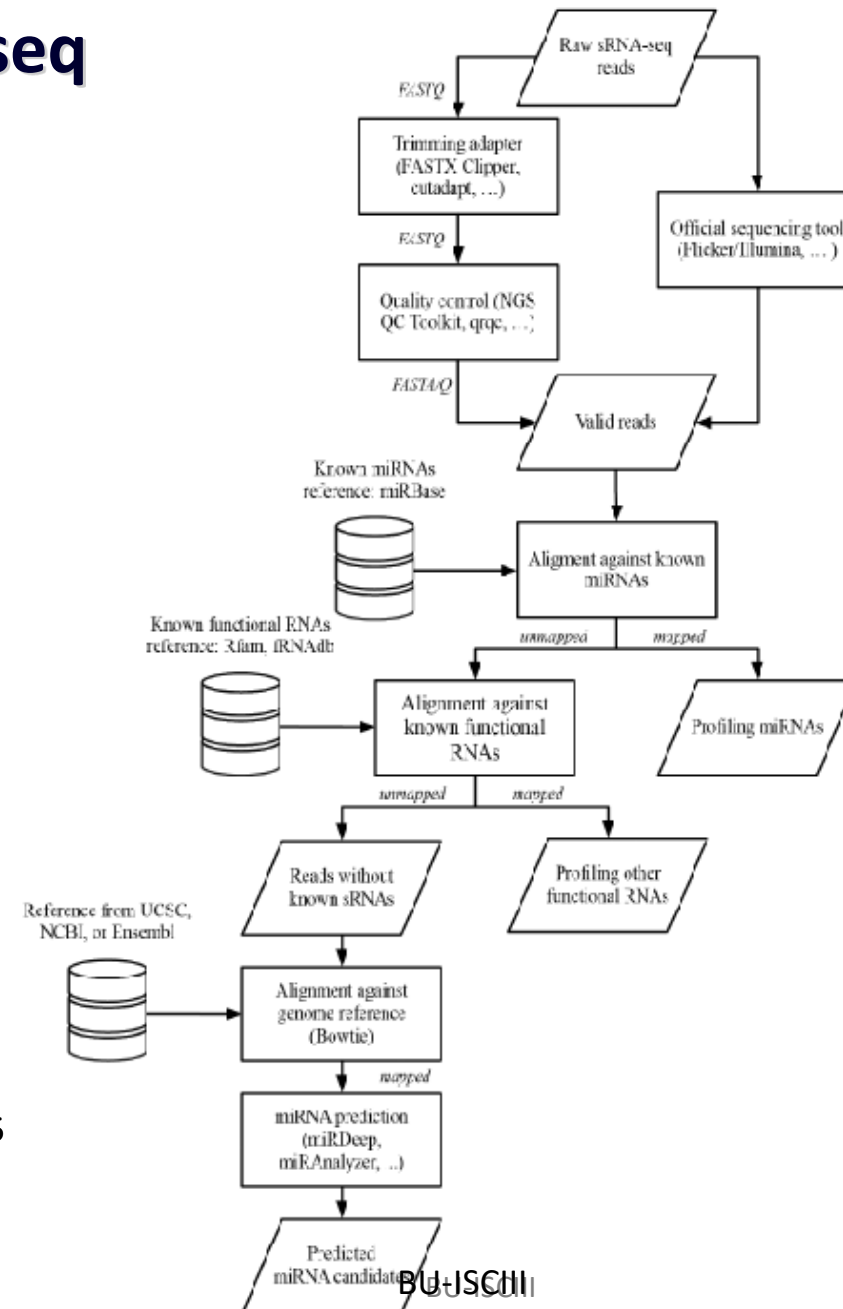
- A diferencia de microarrays no requiere conocimiento de la secuencia del genoma
- Bajo nivel de ruido (las lecturas se mapan correctamente)
- Cuantificación de los transcritos
- Identificación de nuevos transcritos
- Identificación de variaciones (SNPs)
- Disminución progresiva del coste
- Pipelines de análisis disponibles para organismos eucariotas superiores

Strategies of DNA-seq & RNA seq



Lee et al., Transl Cancer Res 2013;2(1):33-45

Strategies of small RNA seq



Lee et al., Transl Cancer Res 2013;2(1):33-45

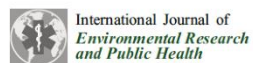
Applications in Human Genetics

- Identification of polymorphisms –SNVs (SNPs, indels)
- Identification of structural variants –SVs (1kb-3Mb) (indels, CNVs, inversions, translocations)
- Gene expression profiling analysis, differential expression – mRNA, SNVs
- Identification and quantification of : smallRNA, miRNA, lncRNA
- De novo transcriptome assembly
- Chip-Seq
- Metil-Seq
- Variant effect annotation

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Clinical Bioinformatics - Precision Medicine

SpainUDP: “Spanish Undiagnosed Rare Diseases Program”



Article

SpainUDP: The Spanish Undiagnosed Rare Diseases Program

Estrella López-Martín ^{1,*}, Beatriz Martínez-Delgado ², Eva Bermejo-Sánchez ¹, Javier Alonso ², the SpainUDP Network [†] and Manuel Posada ¹

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* Correspondence: elopez@isciii.es; Tel: +34-91-822-2911

[†] SpainUDP (Spain) is a member of UDNI (Undiagnosed Diseases Network International) (network's members who are co-authors are listed in Appendix A), 28029 Madrid, Spain.

Received: 12 June 2018; Accepted: 7 August 2018; Published: 14 August 2018



SpainUDP

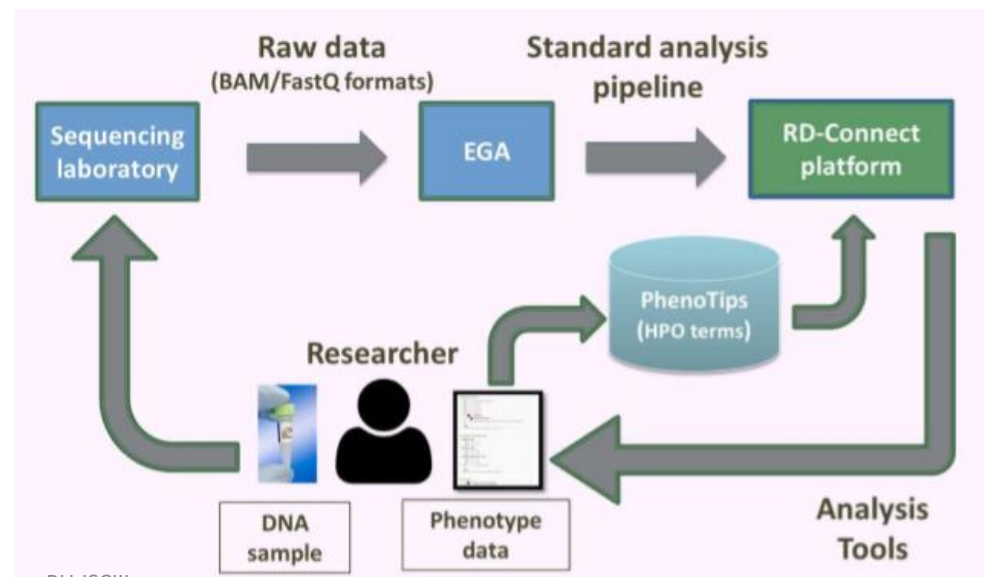
RDConnect

EUROPEAN JOINT PROGRAMME
RARE DISEASES

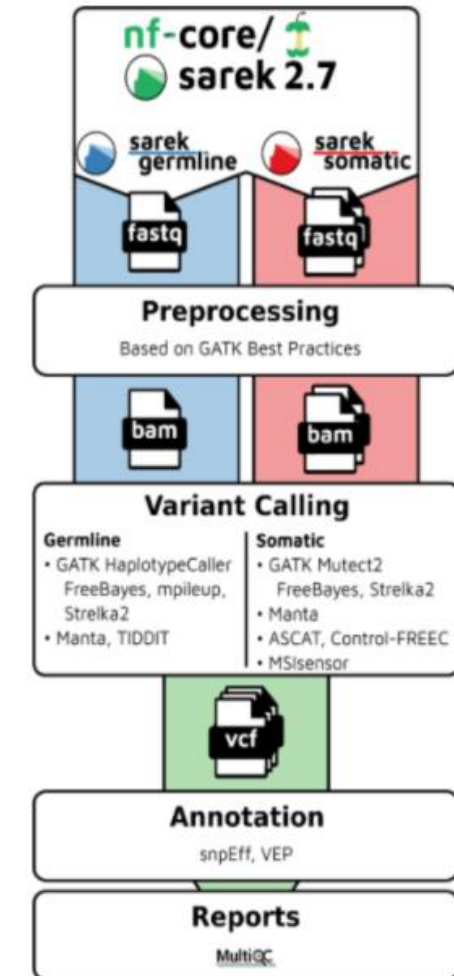
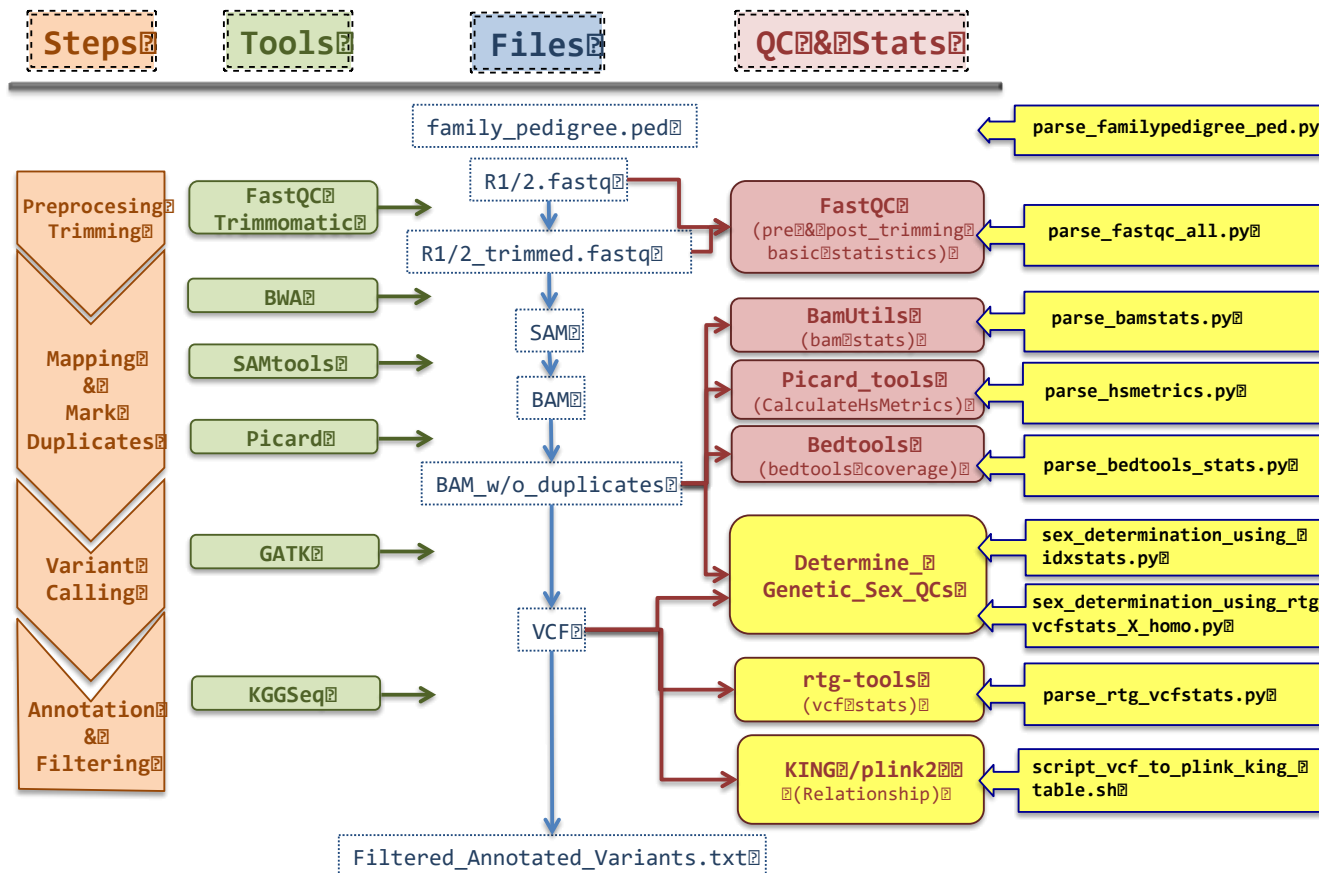
SolveRD

SpainUDP, Manuel Posada

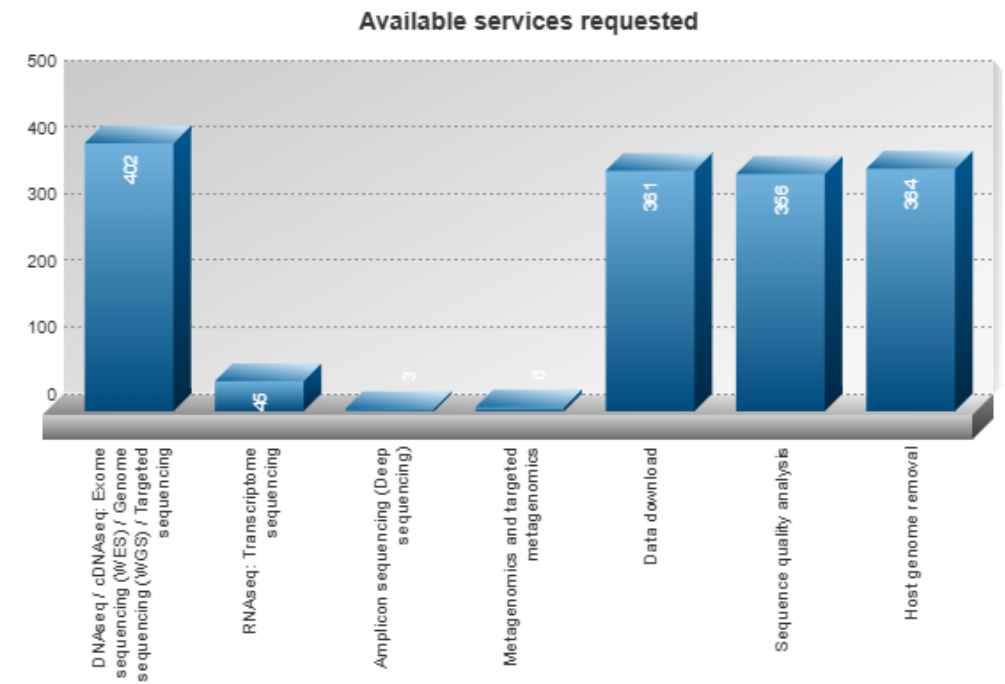
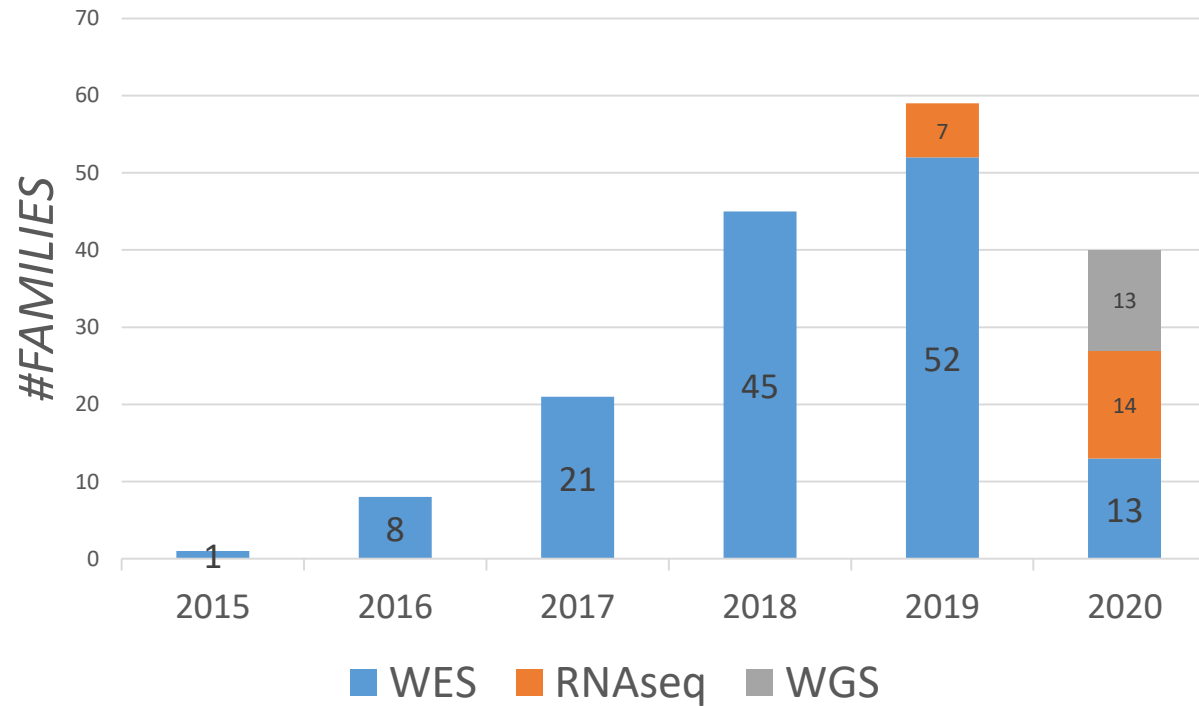
- El programa SpainUDP - IIER se creó con el objetivo de intentar obtener un diagnóstico clínico en pacientes con enfermedades raras sin diagnosticar. Se basa en tres pilares: Estudio individualizado y exhaustivo de cada caso; Uso de técnicas de NGS; Compartir datos a nivel internacional (RD-Connect)
- Se han analizado 120 WES, llegando a un 50% de resolución de casos. Se incorporó en 2020 WGS y RNAseq.



Data analysis: Pipeline



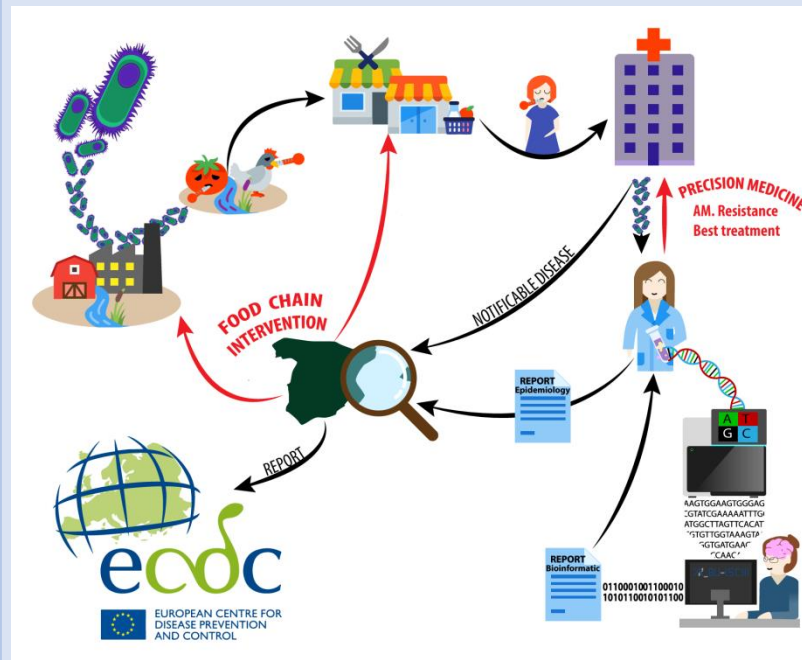
SpainUDP: number of families analyzed



Microbial Genomics Applications

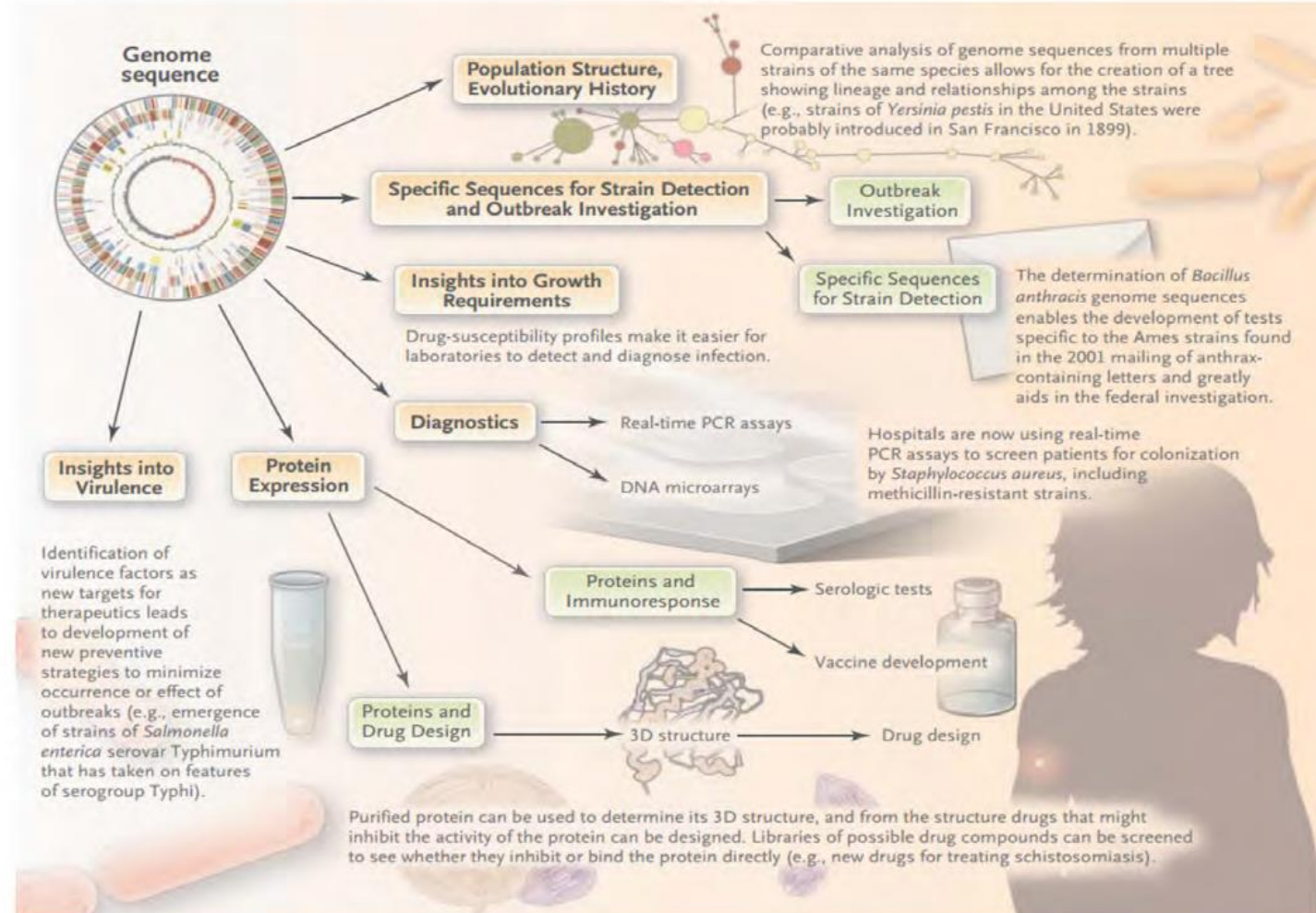
Research - Clinical Bioinformatics - Precision Medicine

CNM Pathogen associated



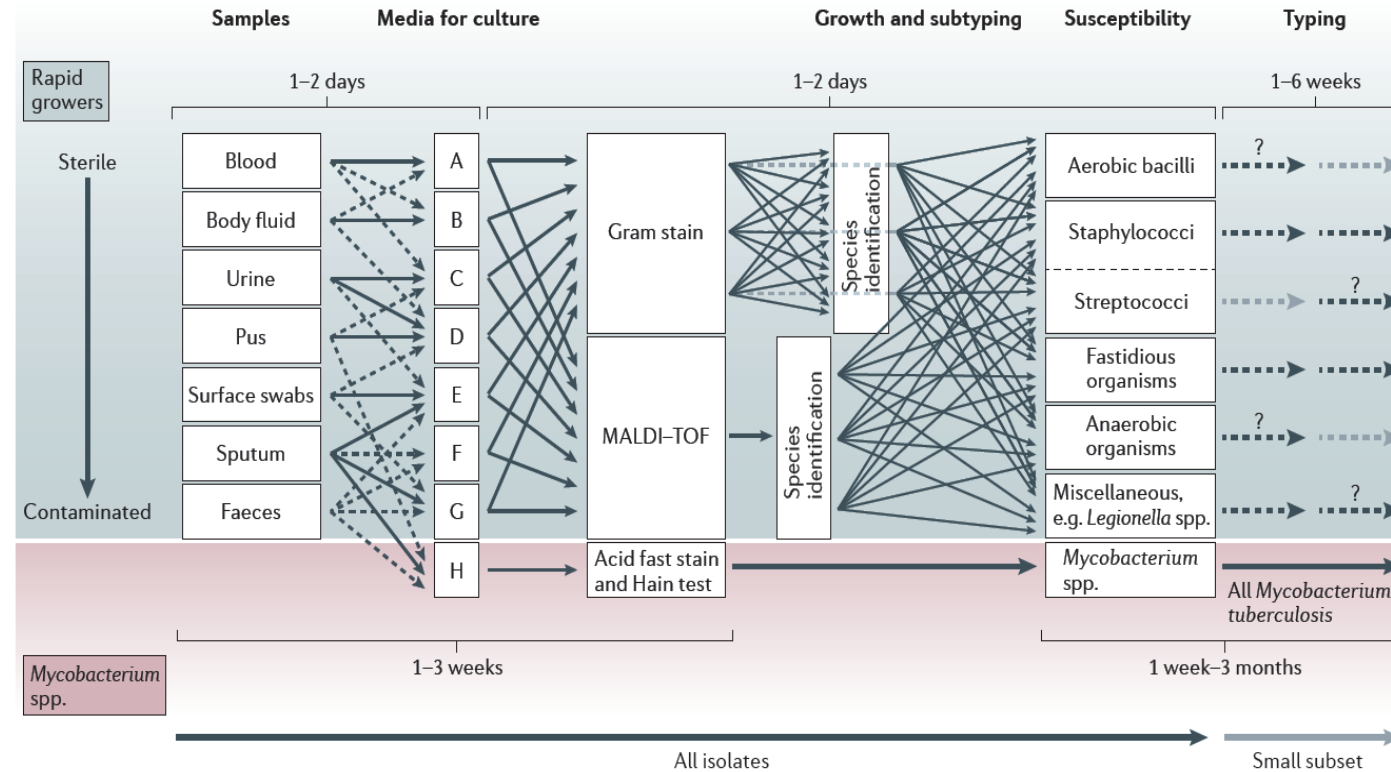
Use of microbial genomics for tool development

Report from The American Academy of Microbiology, 2015



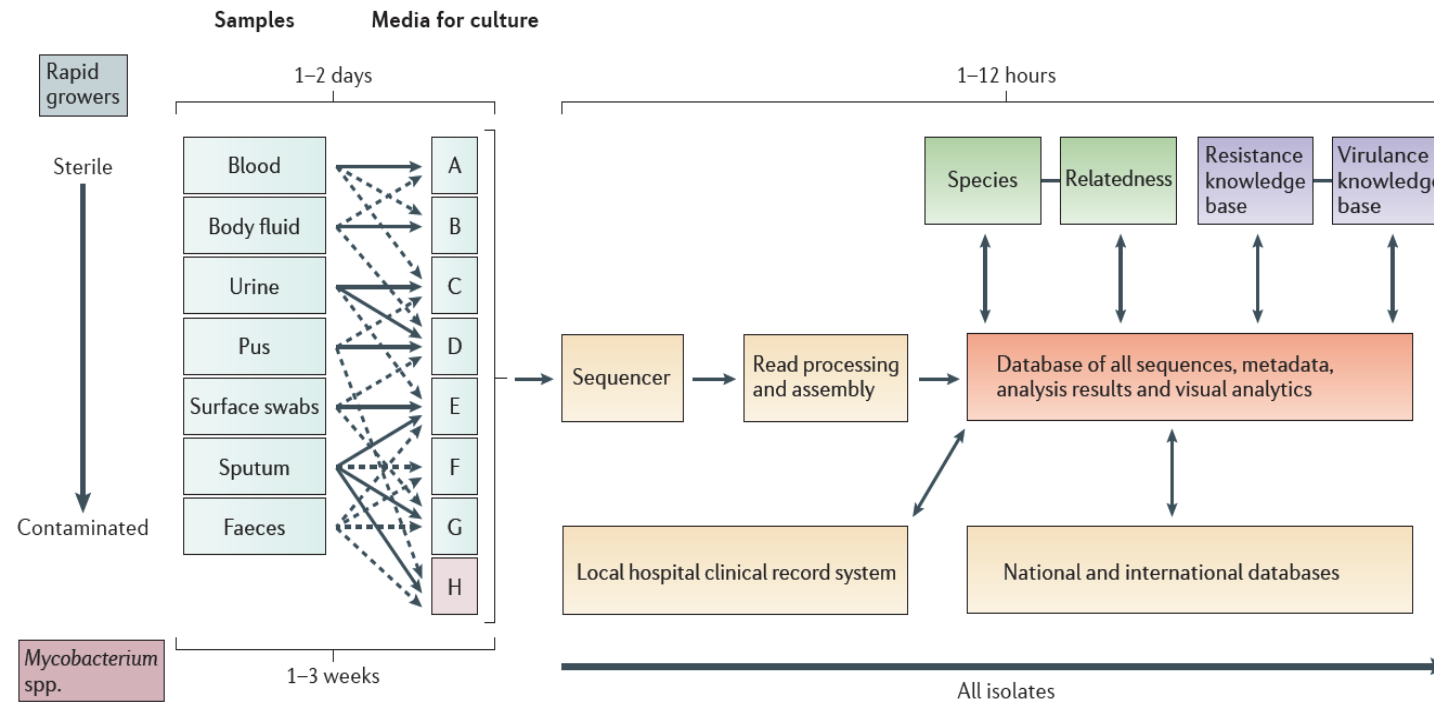
Workflow for processing samples for bacterial pathogens

Didelot et al., Nature Genet Review 2012, 13:601-612



Ongoing developments in DNA-sequencing technologies are likely to affect the diagnosis and monitoring of all pathogens, including viruses, bacteria, fungi and parasites.

The diagnostic and clinical applications of bacterial WGS



Didelot et al., Nature Genet Review 2012, 13:601-612

Foodborne outbreak identification “Crisis del pepino”

2011

Mayo

- 24 Primera muerte en Alemania
- 26 Alemania acusa a los pepinos españoles
- 30 Prohibición de importaciones de verduras de España y Alemania
- 31 Laboratorios alemanes desmienten oficialmente que los pepinos españoles sean el foco de infección

Junio

- 10 Resolución de la crisis

Causado por la toxi-infección de *Escherichia coli* enterohemorrágica (EHEC) (*Escherichia coli* O104:H4)

Muerte: 32 personas en Alemania, 1 Suecia y 1 Francia y 2263 infectados en 12 países de Europa.

Crisis Política y Económica Europa: Alto impacto en la Economía Europea, mayor afectación en la Española

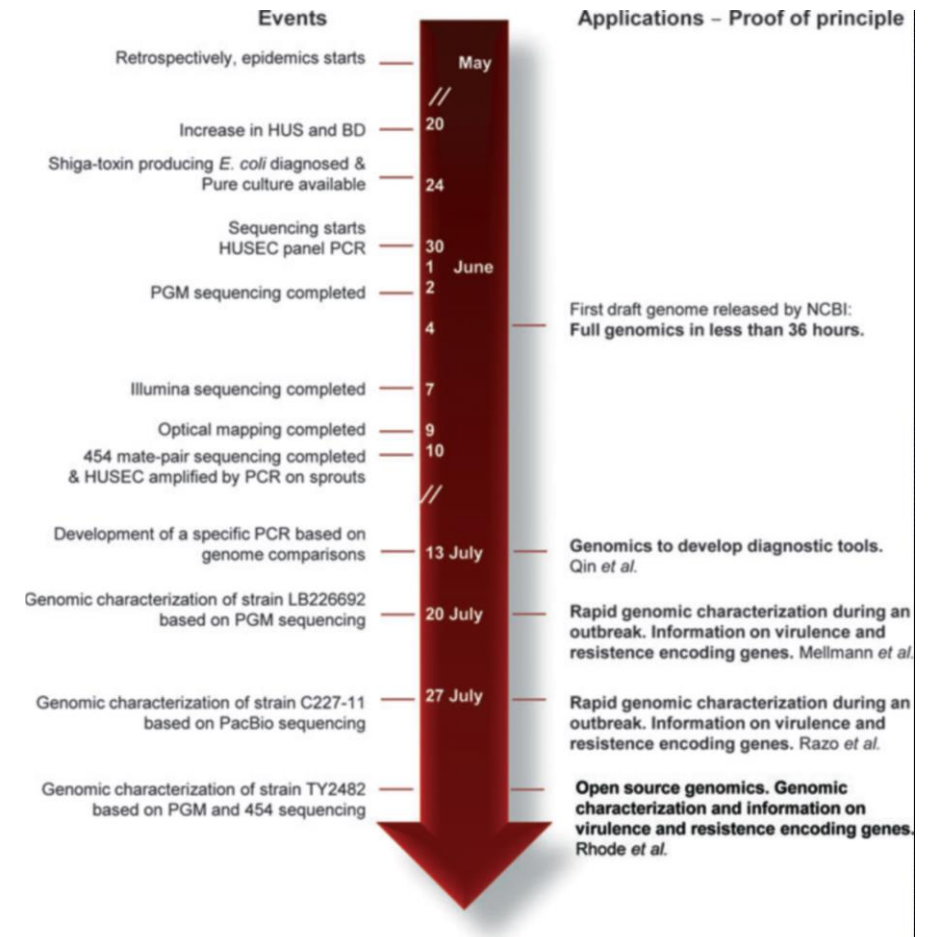
Secuenciación Genoma

华大基因
BGI

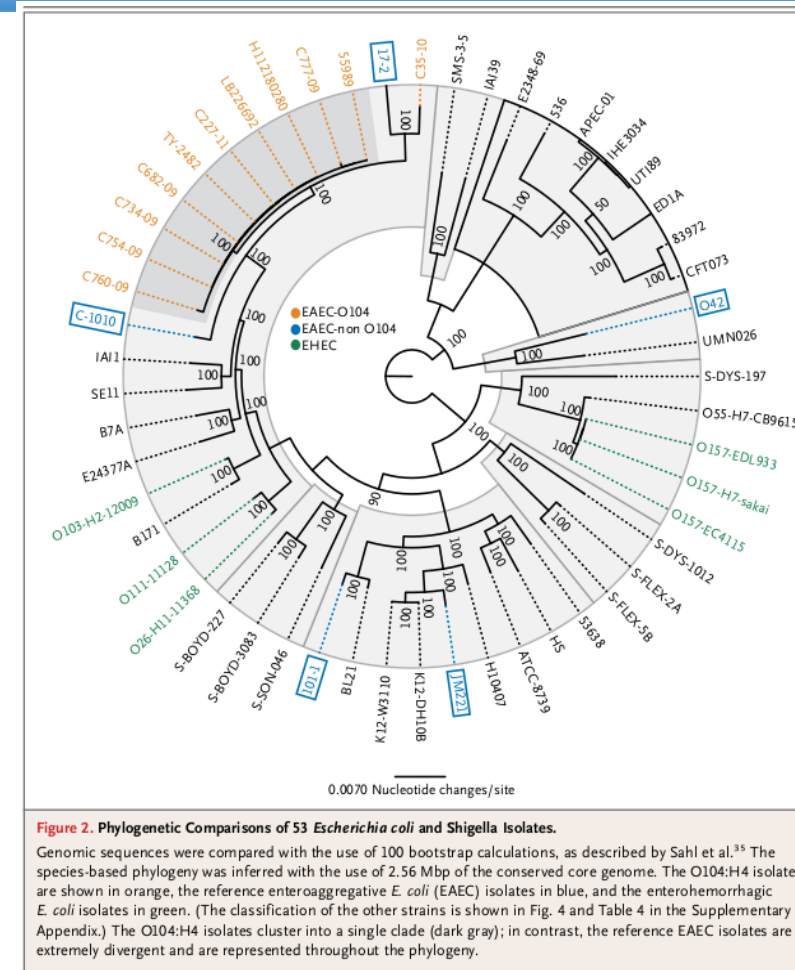
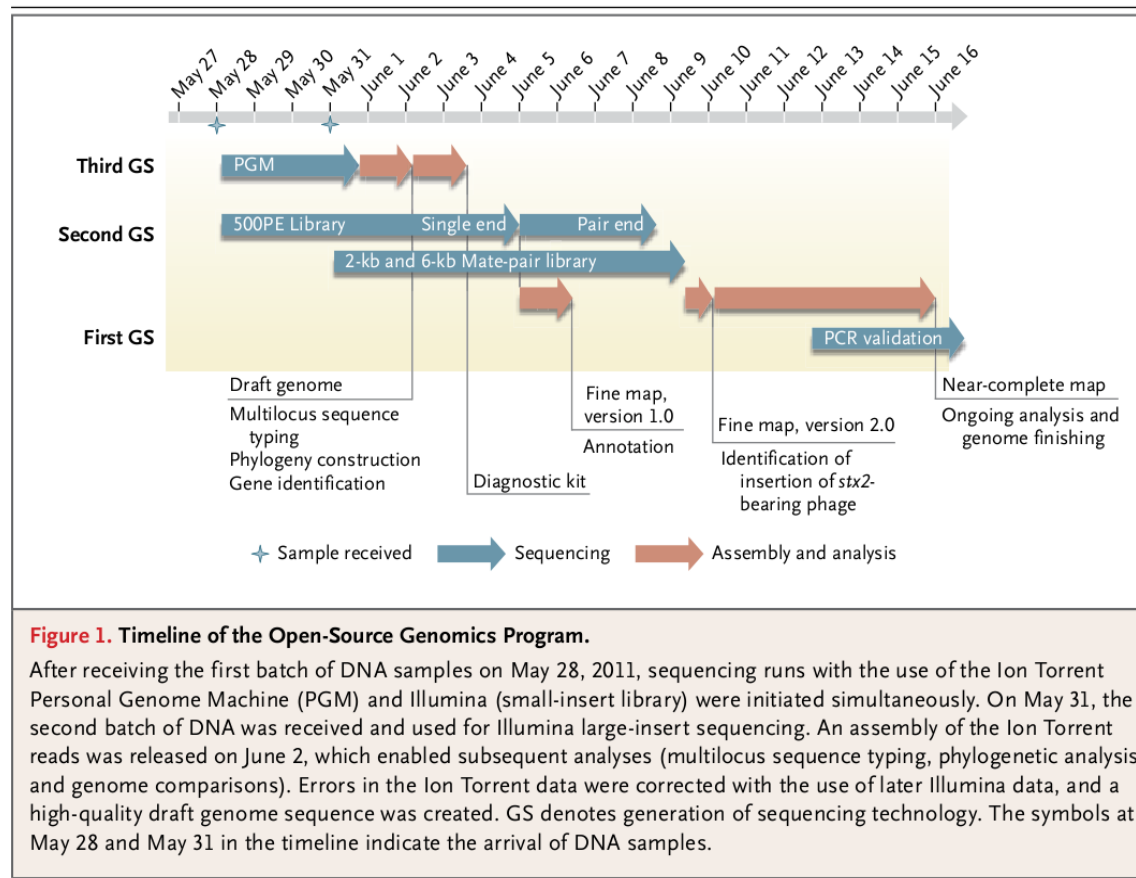


Universitätsklinikum
Hamburg-Eppendorf

The *Escherichia coli* O104:H4 epidemics: event timeline and major outputs



Foodborne outbreak identification “Crisis del pepino”



Andalusian Listeria Outbreak

Actualización de información sobre el brote de intoxicación alimentaria causado por *Listeria monocytogenes*.

Publica: Agencia Española Seguridad alimentaria y Nutrición
Fecha: 29 agosto 2019
Sección: Seguridad Alimentaria

Jueves 29 de agosto de 2019, 12.00 horas

ACTUALIZACIÓN EN RELACIÓN CON LA DISTRIBUCIÓN DE PRODUCTOS RELACIONADOS CON LA ALERTA.

La Agencia Española de Seguridad Alimentaria y Nutrición (AESAN) recomienda a las personas que tengan en su domicilio algún producto de la marca "La Mechá" se abstengan de consumirlo. Si se dispone del producto se debe devolver al punto de compra y, de no ser posible, desecharlo.

Brote de listeriosis: sube el número d afectados y se apunta a la falta de higiene en la carne como causa

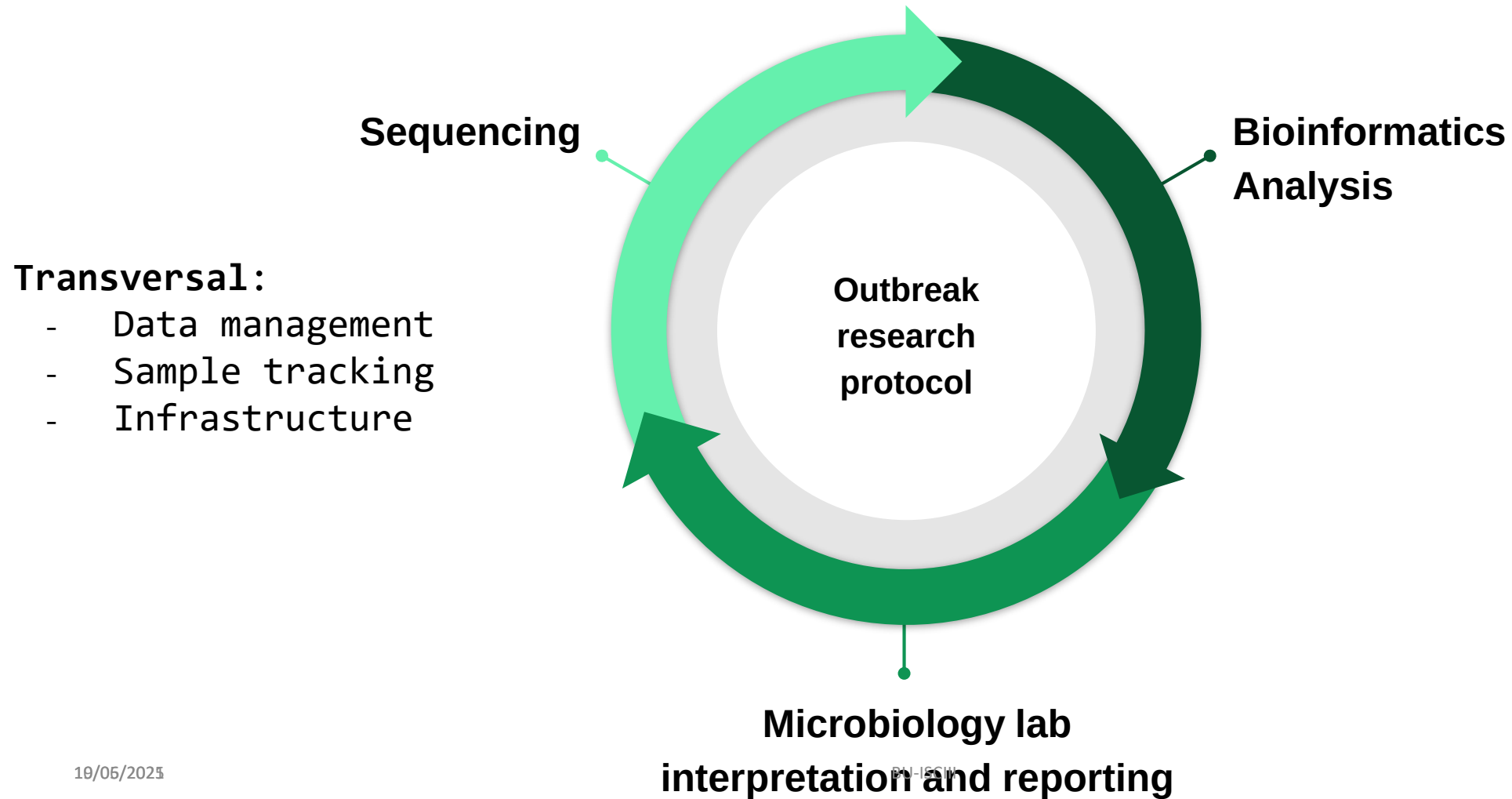
EFE 25.08.2019



- Tres nuevos casos, en Sevilla y Cádiz, dejan el número de personas afectadas en Andalucía en 192.
- [La carne con listeria de la marca blanca se vendió en los municipios de Sevilla.](#)
- La empresa que vendió la marca blanca de Magrudis dice que cumple los protocolos.

- Meat "La Mechá". Margulis S.L.
- 250 cases related.
- Meat "'La Montanera del Sur".INCARYBE S.L", suspicion. (Cádiz)
- Meat "Sabores de Paterna" (Málaga)

Andalusian Listeria Outbreak

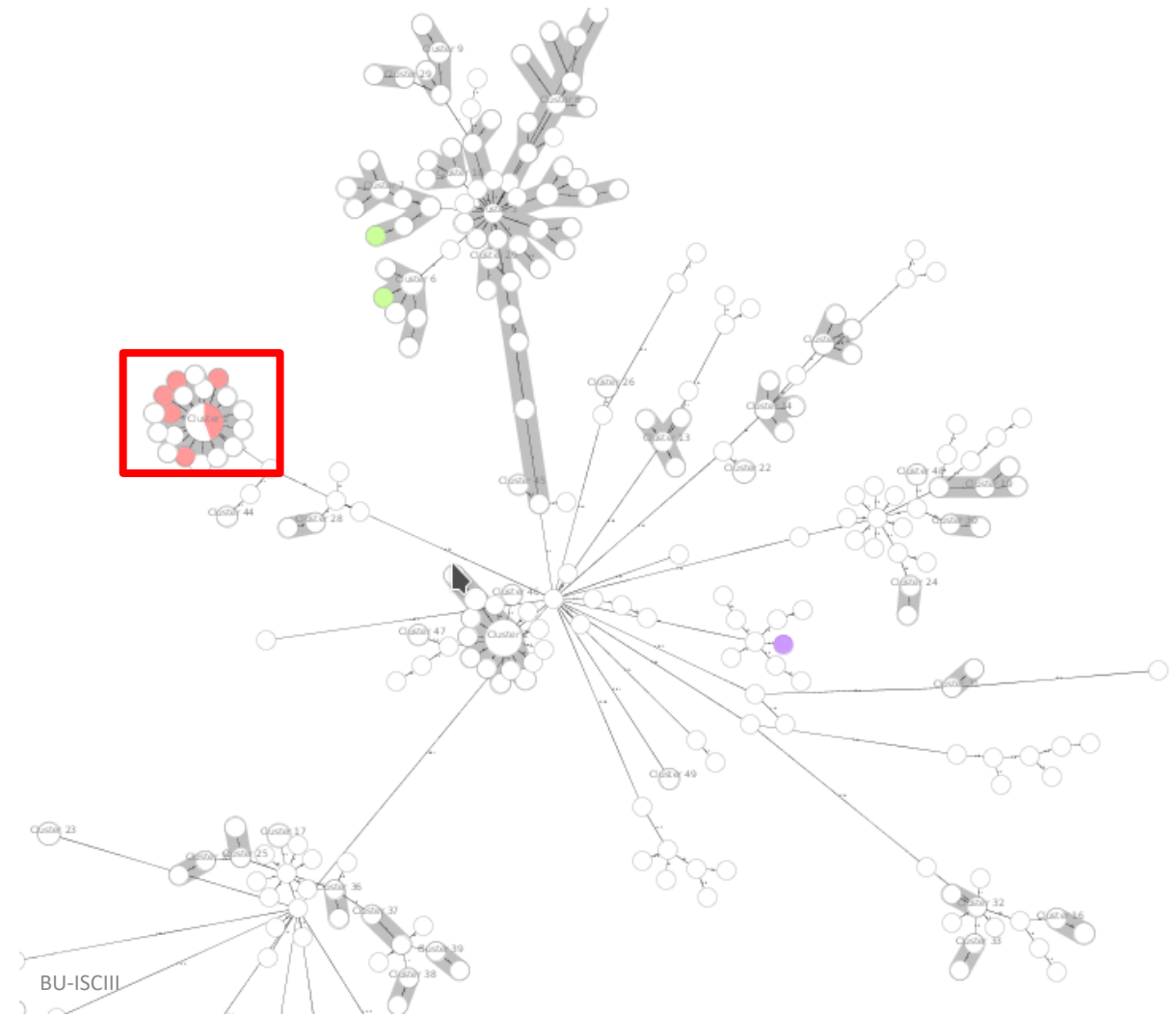


Andalusian Listeria Outbreak

- 625 listeria samples already sequenced
- 258 suspected to be related to the outbreak (mid august to mid september)

Results:

- 233 related to the outbreak, confirmed to be caused by the meat “La Mechá”
- 25 sporadic cases not related to the outbreak.



Pathogen discovery: new virus – SARS-CoV-2

Deep Meta-Transcriptomic Sequencing



bronchoalveolar lavage fluid (BALF)



Meta-transcriptomic library

2x150 MiniSeq 56,565,928 sequences reads

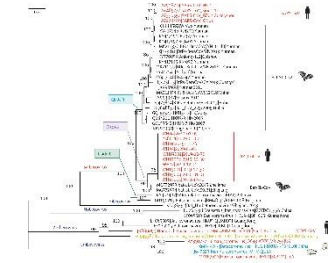


De novo-assembled - Megahit

384,096 Contigs

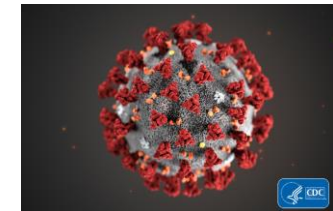
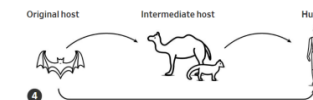
Screened for potential aetiological agents

The longest 30,474 nt



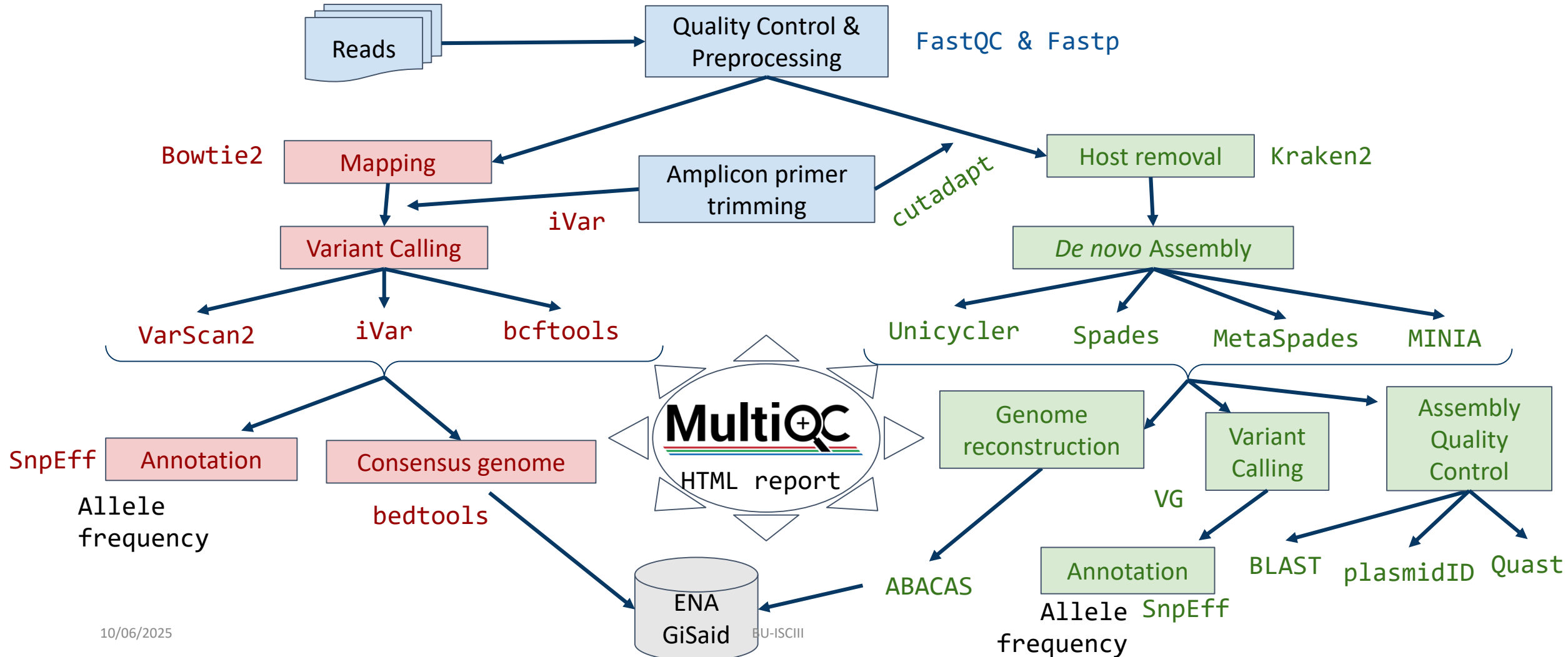
89.1% identity

Closely related to a bat SARS-like coronavirus



Wu et al., Nature 2020

Viralrecon



Surveillance-based reporting

Virussequenc	sample	totalreac	readshostR	readsho	%readsho	readsviru	%readsviru	unmappedreac	%unmapedreac	meanDPcoverageviru	Coverage>10x(%)	Variantsinconsensusx1	MissenseVarianti	%Ns1C	Lineages
NC_045512.2	211524	2040680	24760	49520	2.43	1980480	97.05	10680	0.52	4607.76	99.79	41	28	0,17	B.1.1.7
NC_045512.2	210710-2	2166652	12091	24182	1.12	2140652	98.80	1818	0.08	5331.12	99.77	40	25	0,18	B.1.1.7
NC_045512.2	211496	2092444	2299	4598	0.22	2081982	99.50	5864	0.28	4684.85	99.55	35	23	0,22	B.1.1.7
NC_045512.2	211429	1739402	1643	3286	0.19	1730009	99.46	6107	0.35	3891.88	98.84	46	27	0,24	B.1.1.7
NC_045512.2	211507	1394744	1500	3000	0.22	1386515	99.41	5229	0.37	3438.01	99.05	39	26	0,25	B.1.1.7
NC_045512.2	211517	2284388	6870	13740	0.60	2263829	99.10	6819	0.30	4882.96	99.66	40	26	0,26	B.1.1.7
NC_045512.2	211405	1946156	19779	39558	2.03	1900227	97.64	6371	0.33	4174.87	99.57	37	25	0,32	B.1.1.7
NC_045512.2	211497	1823806	104	208	0.01	1821435	99.87	2163	0.12	4889.73	99.65	39	25	0,33	B.1.1.7
NC_045512.2	211430	1859150	378	756	0.04	1853758	99.71	4636	0.25	4475.72	99.66	20	7	0,34	B.1.177
NC_045512.2	211498	1974394	713	1426	0.07	1963930	99.47	9038	0.46	4331.72	99.63	38	25	0,34	B.1.1.7
NC_045512.2	211506	2054730	28336	56672	2.76	1987951	96.75	10107	0.49	3822.90	99.15	41	27	0,49	B.1.1.7
NC_045512.2	211516	2240664	446	892	0.04	2232598	99.64	7174	0.32	4427.42	99.16	19	8	0,61	B.1.177
NC_045512.2	211495	2245802	4263	8526	0.38	2218628	98.79	18648	0.83	3620.33	98.60	35	23	0,62	B.1.1.7
NC_045512.2	211513	2191920	1071	2142	0.10	2183372	99.61	6406	0.29	4394.22	99.17	18	8	0,66	B.1.177
NC_045512.2	211489	1995358	619	1238	0.06	1990170	99.74	3950	0.20	4938.41	99.14	39	27	0,7	B.1.1.7
NC_045512.2	211512	2032422	11430	22860	1.12	1985067	97.67	24495	1.21	3070.94	98.21	39	26	0,7	B.1.1.7
NC_045512.2	211419	2143186	2827	5654	0.26	2125398	99.17	12134	0.57	4100.12	99.08	41	24	0,77	B.1.1.7
NC_045512.2	211509	1905366	1412	2824	0.15	1895839	99.50	6703	0.35	4146.72	98.79	40	26	0,78	B.1.1.7
NC_045512.2	211411	2192376	1420	2840	0.13	2180537	99.46	8999	0.41	3849.04	98.07	29	18	0,96	B.1.351
NC_045512.2	211519	2406852	382	764	0.03	2390245	99.31	15843	0.66	3833.59	98.42	27	12	0,99	B.1.177
NC_045512.2	210790-2	2042452	32448	64896	3.18	1928075	94.40	49481	2.42	3288.34	98.38	32	18	1,08	B.1.160
NC_045512.2	211428	2055490	33405	66810	3.25	1975531	96.11	13149	0.64	3604.64	97.95	37	23	1,22	B.1.1.7
NC_045512.2	211511	1989742	549	1098	0.06	1980788	99.55	7856	0.39	4077.32	98.31	38	26	1,35	B.1.1.7
NC_045512.2	211425	1761330	11924	23848	1.35	1714479	97.34	23003	1.31	2725.52	97.38	39	26	1,55	B.1.1.7
NC_045512.2	211508	1711536	547	1094	0.06	1701951	99.44	8491	0.50	3551.33	98.30	20	13	1,57	C.4
NC_045512.2	211518	2215498	22494	44988	2.03	2158117	97.41	12393	0.56	3630.94	98.03	39	26	1,58	B.1.1.7

One Health approach, infectious diseases could be better controlled and prevented



Spanish National Microbiology Center (CNM)

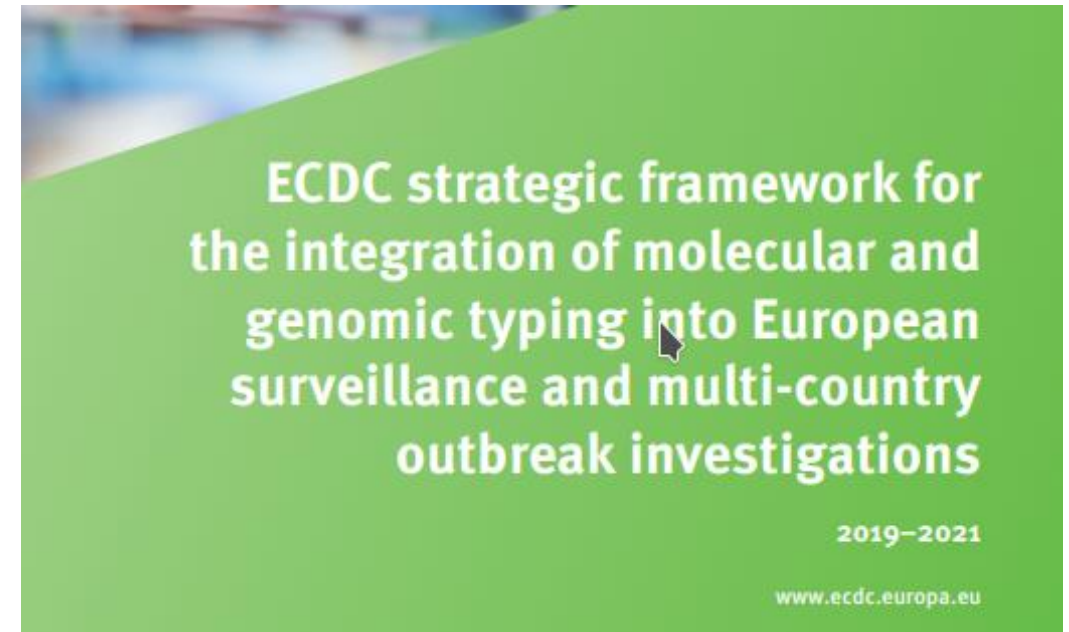


Mission: Provide support to the National Health System and the different Spanish Regions in the diagnosis and control of infectious diseases. In order to fulfill this mission it acts as Reference center offering a series of scientific activities:

- Diagnosis
- **Surveillance** →
- Infectious diseases research
- Training

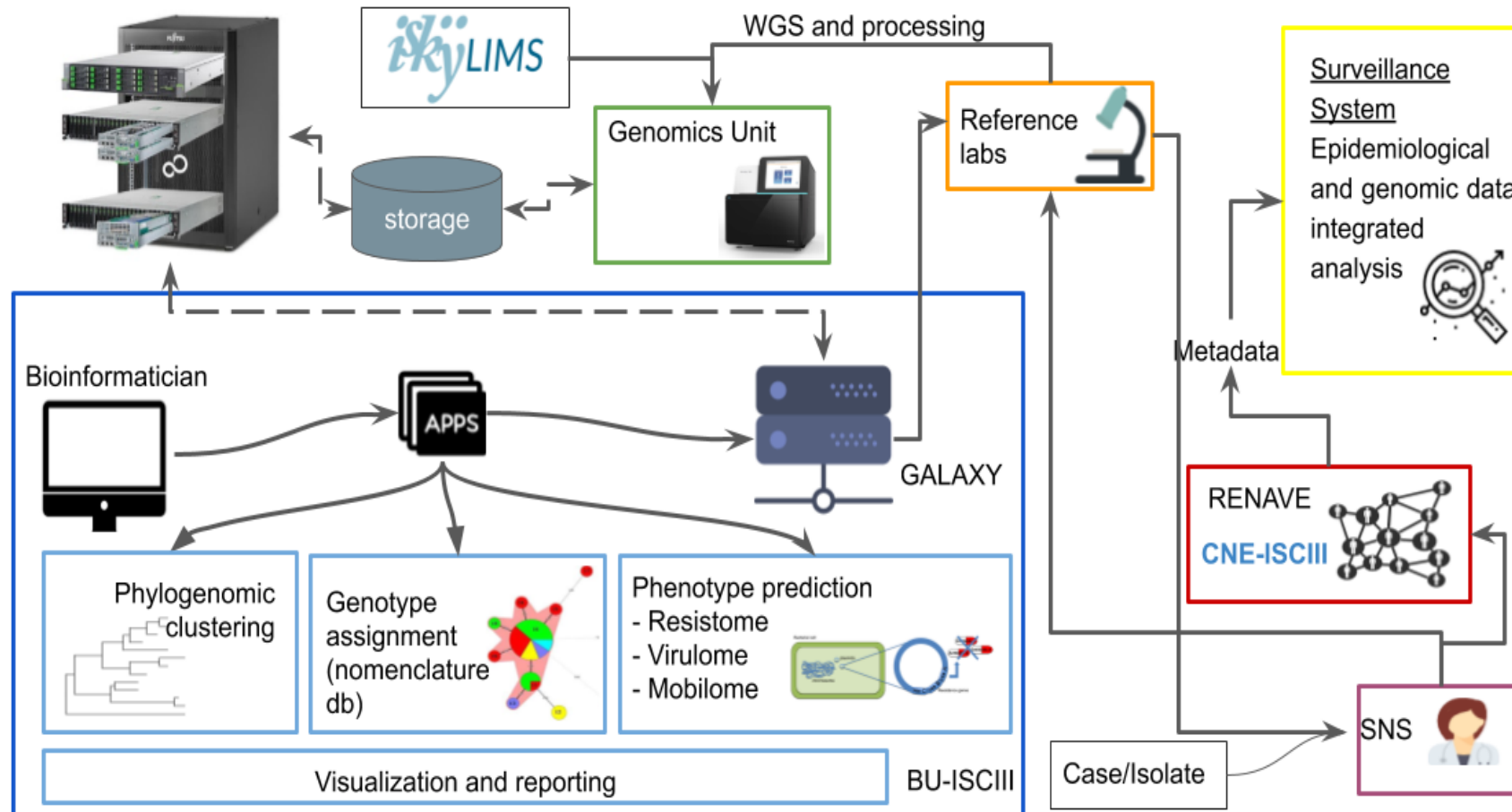
Outbreak research:
Molecular source
detection

ECDC roadmap and international commitment



- **Operationalisation of EU-wide WGS-based surveillance systems in the near term:** ²⁰¹⁸ start implementation of WGS-based surveillance for *Listeria monocytogenes*, *Neisseria meningitidis*, Carbapenemase-producing *Enterobacteriaceae* and antibiotic-resistant *Neisseria gonorrhoeae*;

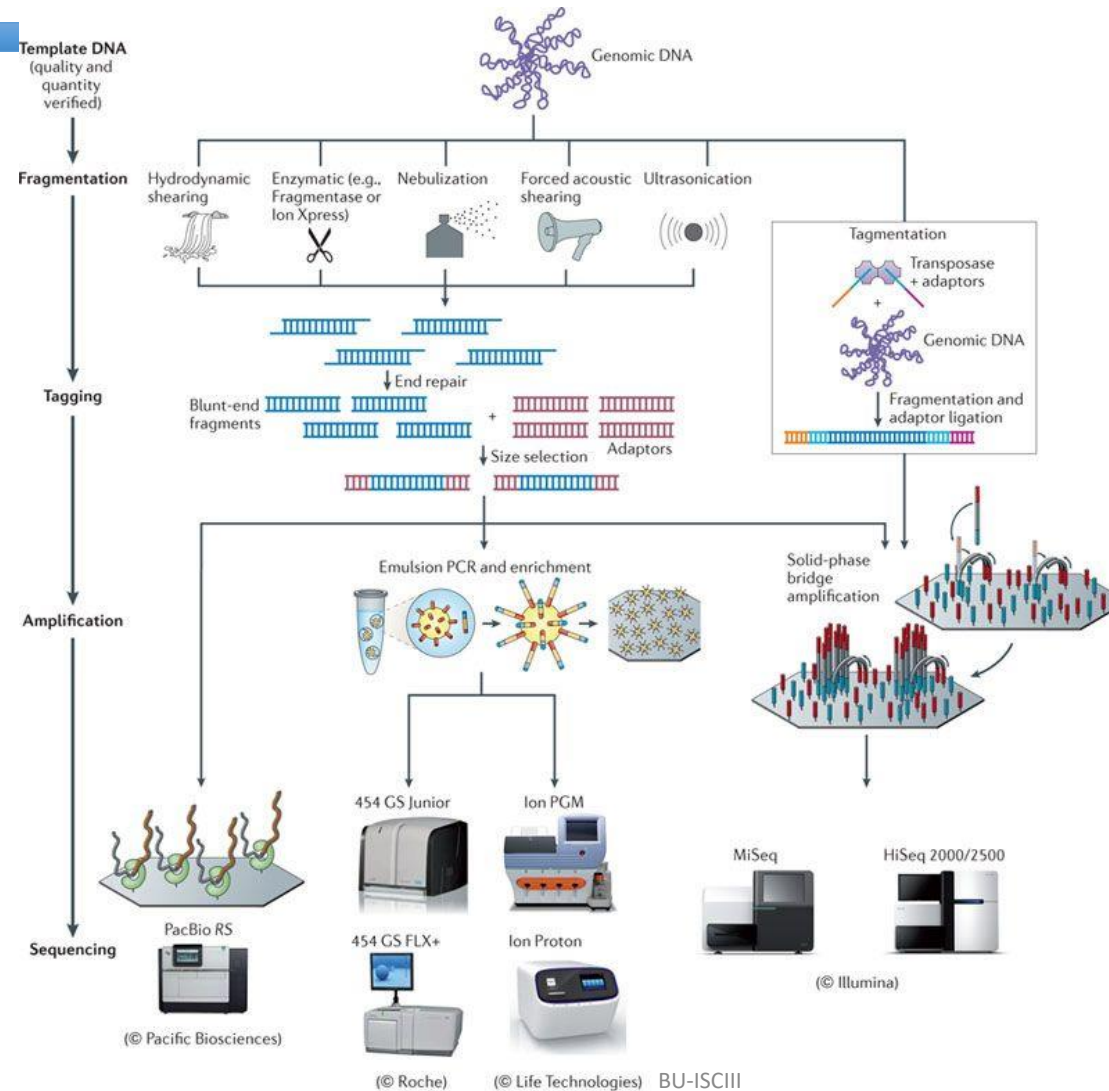
Bioinformatics Unit preparedness



- Infrastructure
- Data
- Sample tracking
- Sharing and intercommunication
- Bioinformatics analysis

Adapted and extended from Expert opinion on whole genome sequencing for public health surveillance. ECDC

High-throughput sequencing platforms



Loman et al, 2012

PREPARACIÓN LIBRERÍA, estrategias

SECUENCIACIÓN GENOMA, EXOMA, TRANSCRIPTOMA

1. Sin amplificación
2. Amplificación con PCR
3. Sondas captura

- Tamaño de fragmento
- Longitud de la lectura
- Single o Paired-end
- Número de bases por muestra
- Profundidad de cobertura x

SECUENCIACIÓN GENOMAS

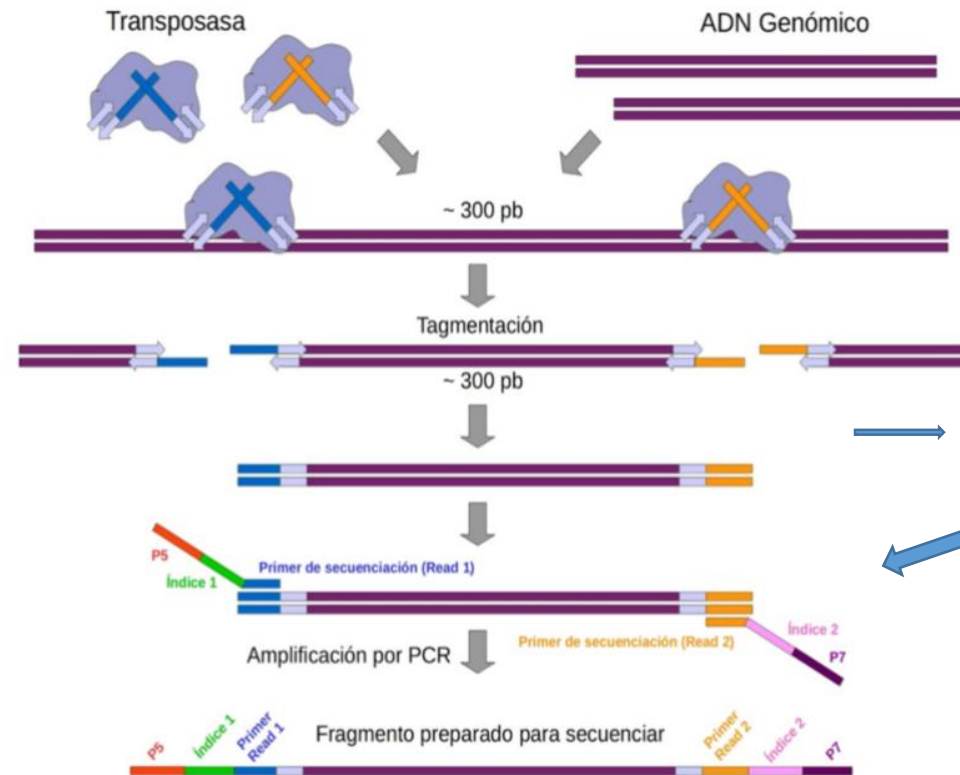
1. Metagenómica

IDENTIFICACIÓN MICROORGANISMOS

1. Metataxonomía

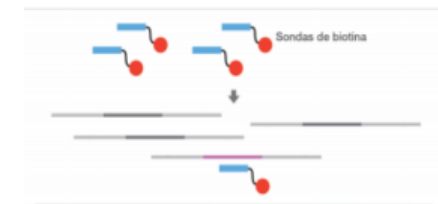
PREPARACIÓN LIBRERÍA

ENZIMÁTICA
FÍSICA



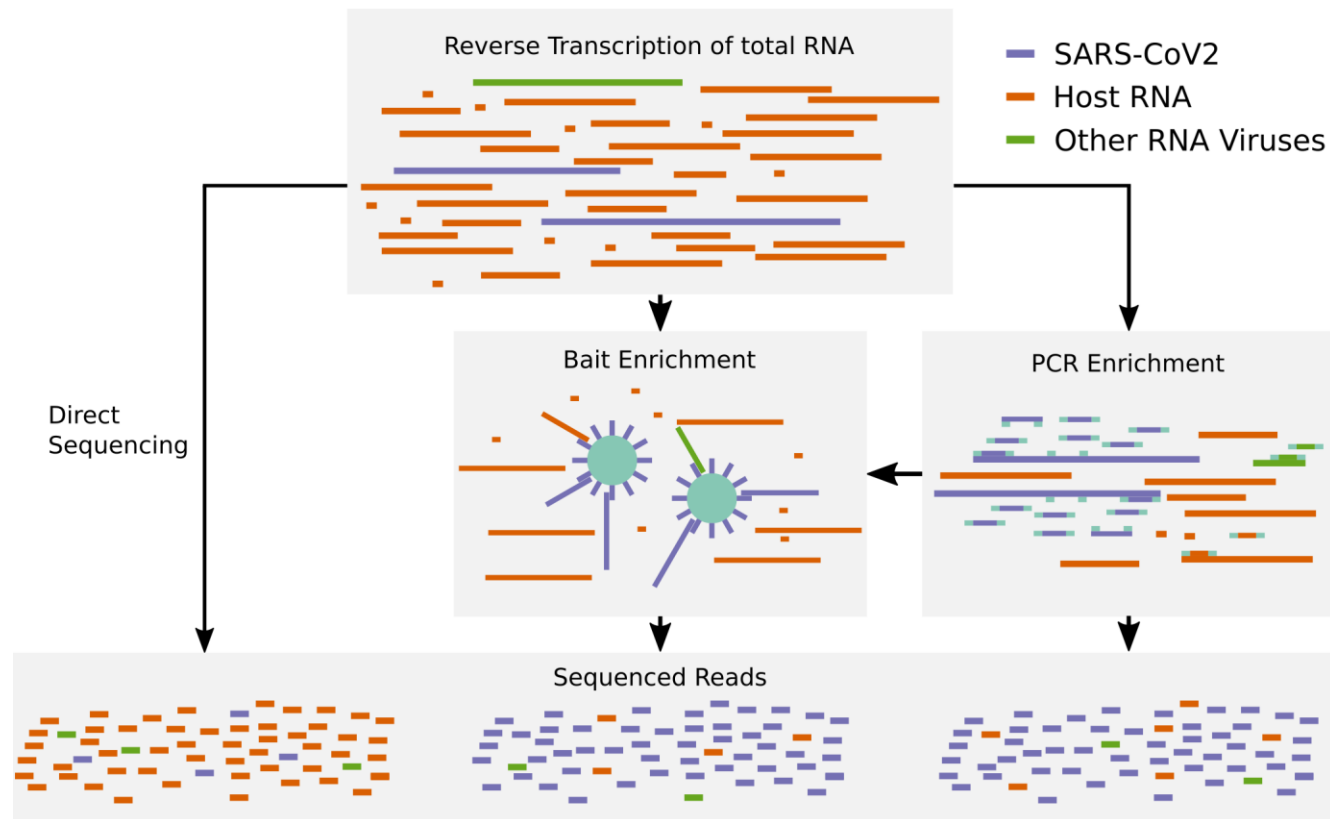
RNA -> cDNA

ENRIQUECIMIENTO:
PCR
CAPTURA SONDAS

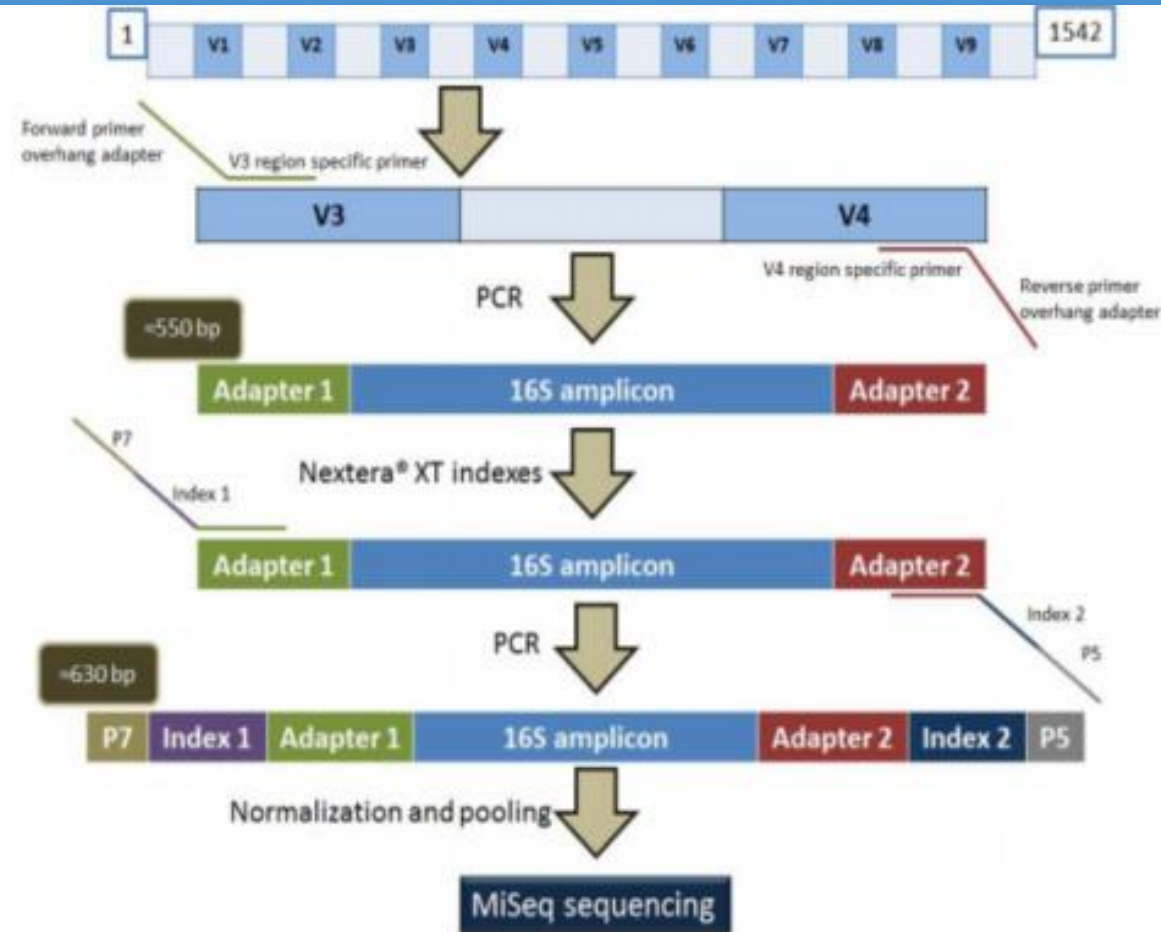


Adaptado de Guia Práctica Genómica https://www.uv.es/varnau/GM_Cap%C3%ADtulo_2.pdf

PREPARACIÓN LIBRERÍA



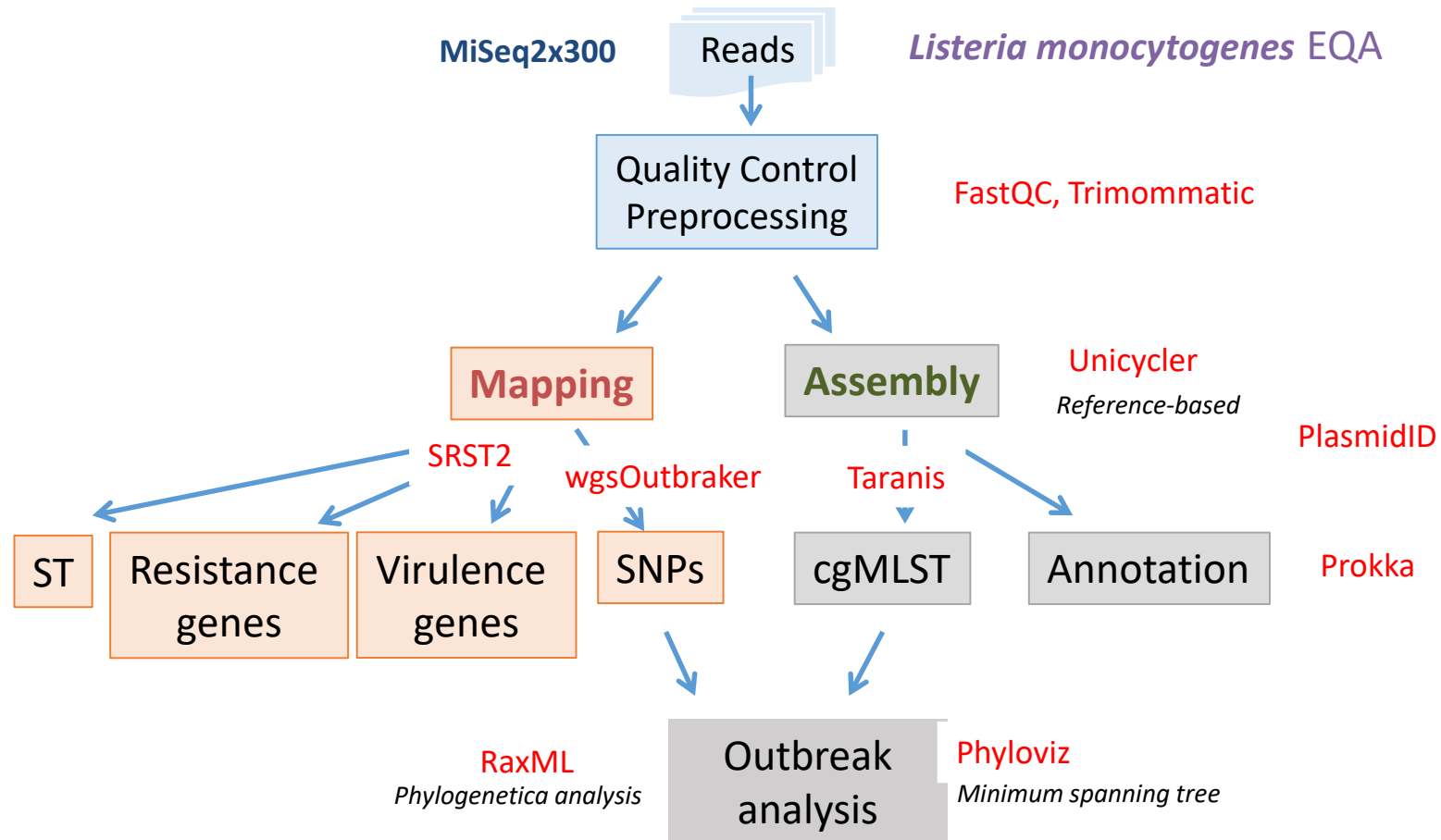
PREPARACIÓN LIBRERÍA, rRNA 16S, caracterización microbiota



Bioinformatics analysis in microbial genomics

- SPECIE IDENTIFICATION
 - WGS - Kmers analysis
 - TARGETED METAGENOMIC, rRNA - MICROBIOTA
- ASSEMBLY GENOME
 - de NOVO or REFERENCE -BASED
 - cgMLST, wgMLST - MINIMUM SPANING TREE
 - METAGENOMIC - HOMOLOGY -BASED
- VARIANT CALLING
 - REFERENCE GENOME SELECTION
 - HAPLOYD GENOME
 - LOW FREQUENCY VARIANT - QUASISPECIES
 - SNPs MATRIX - PHYLOGENETIC ANALYSIS
- STRUCTURAL AND FUNCTIONAL ANNOTATION
 - RESISTOME, VIRULOME, SEQUENCE-TYPE

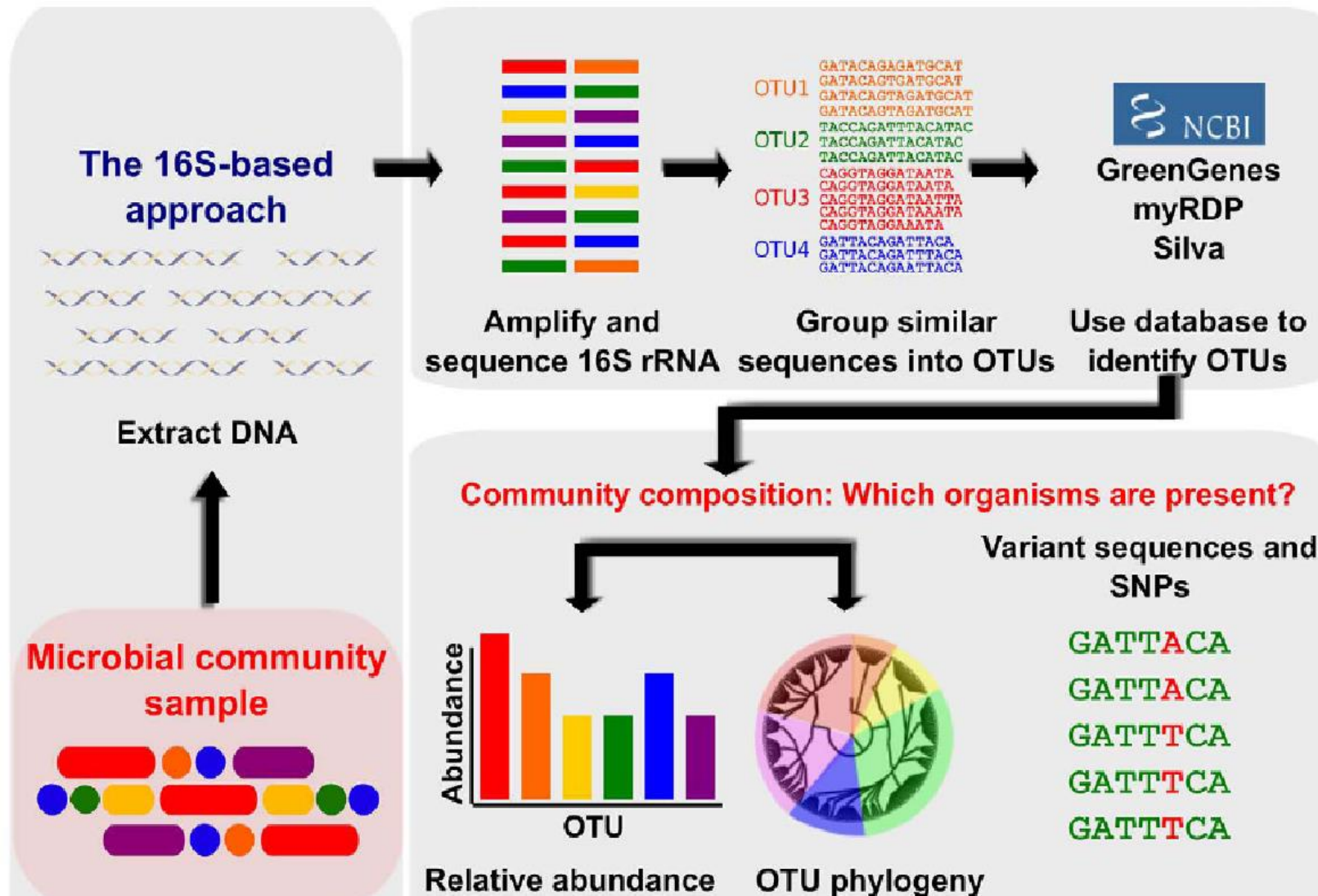
Workflow example



Metataxonomics or targeted metagenomics vs Metagenomics (16S - targeted vs Shotgun)

	Metagenetics	Metagenomics
Amplified sequence	Marker regions	Whole genome
Computing time	Usually short	Usually long
Taxonomic composition	Yes	Yes
New pathogen detection	No	Yes
Genome coverage information	No	Yes

Metataxonomics

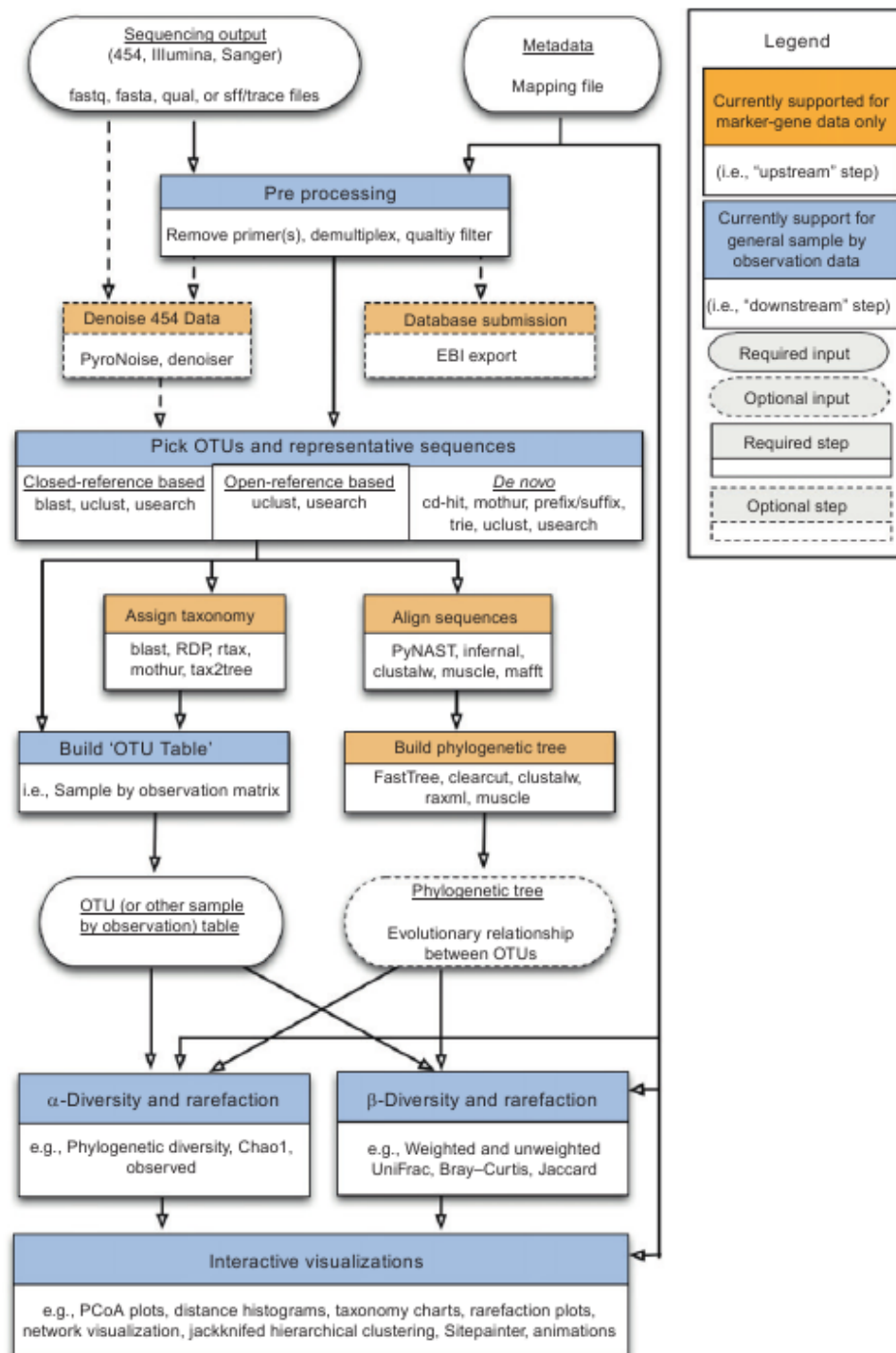


Targeted metagenomics

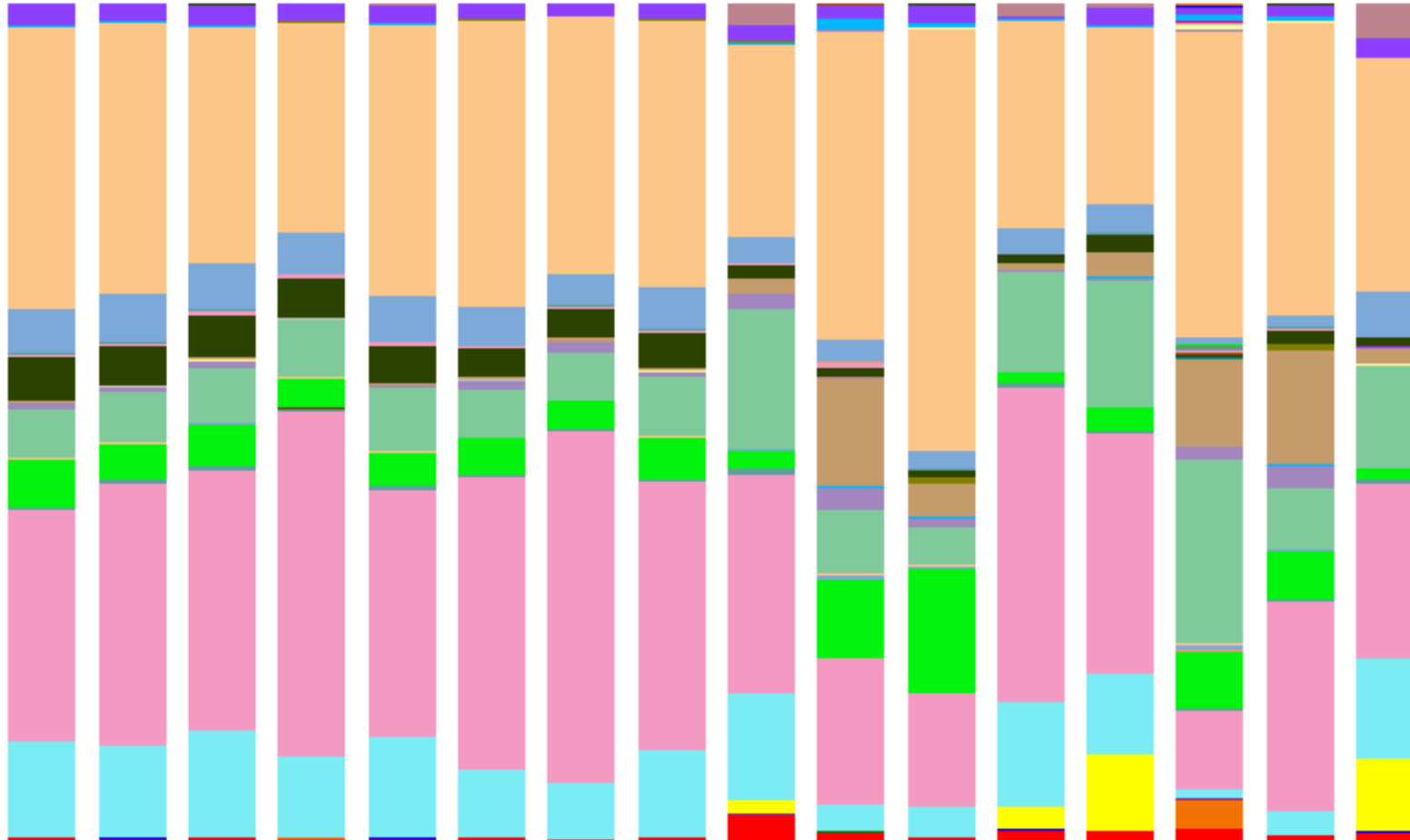
QIIME WORKFLOW

Integrated pipeline of
third-party tools

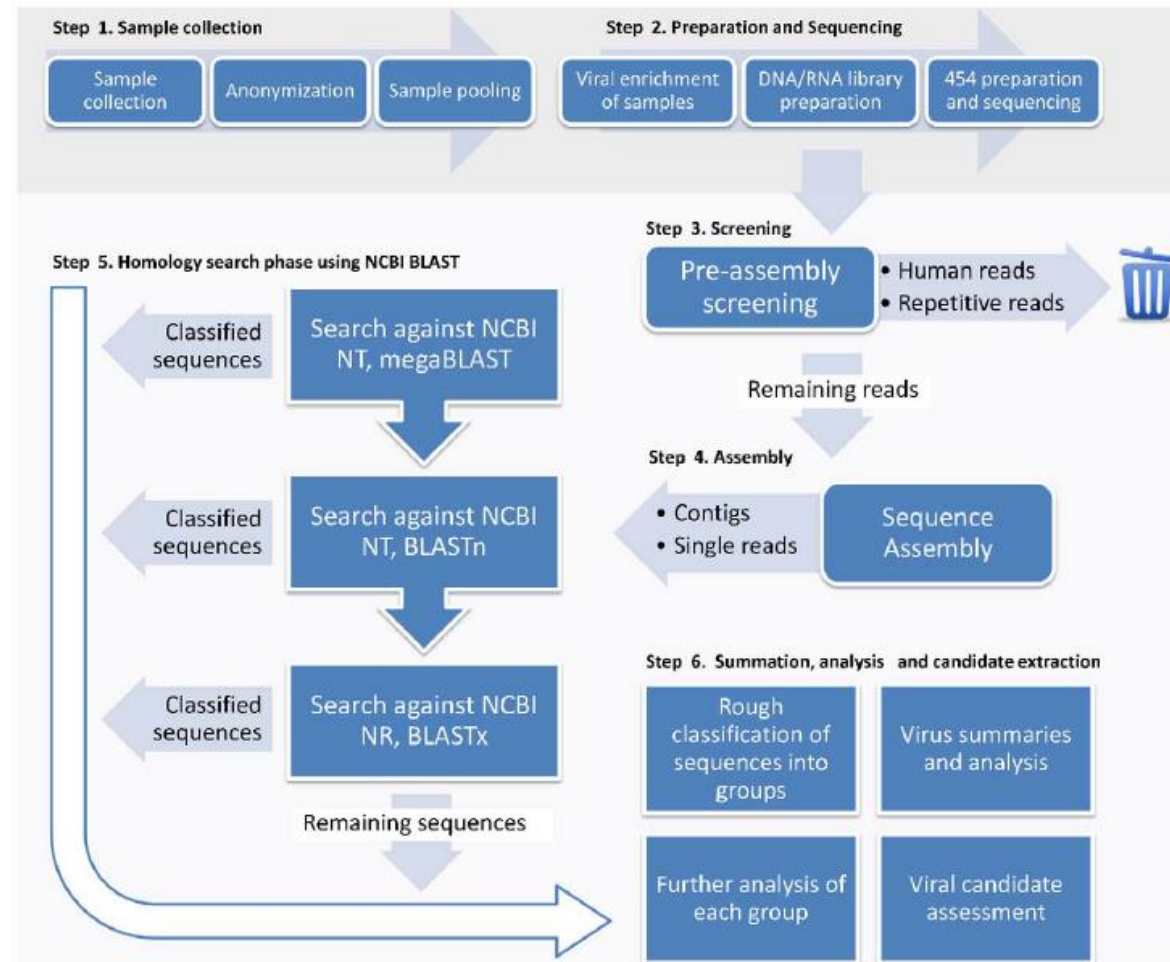
Navas-Molina et al., Methods in
enzymology 2013, 531: 371-439



Taxonomy summary (i.e. phylum level)

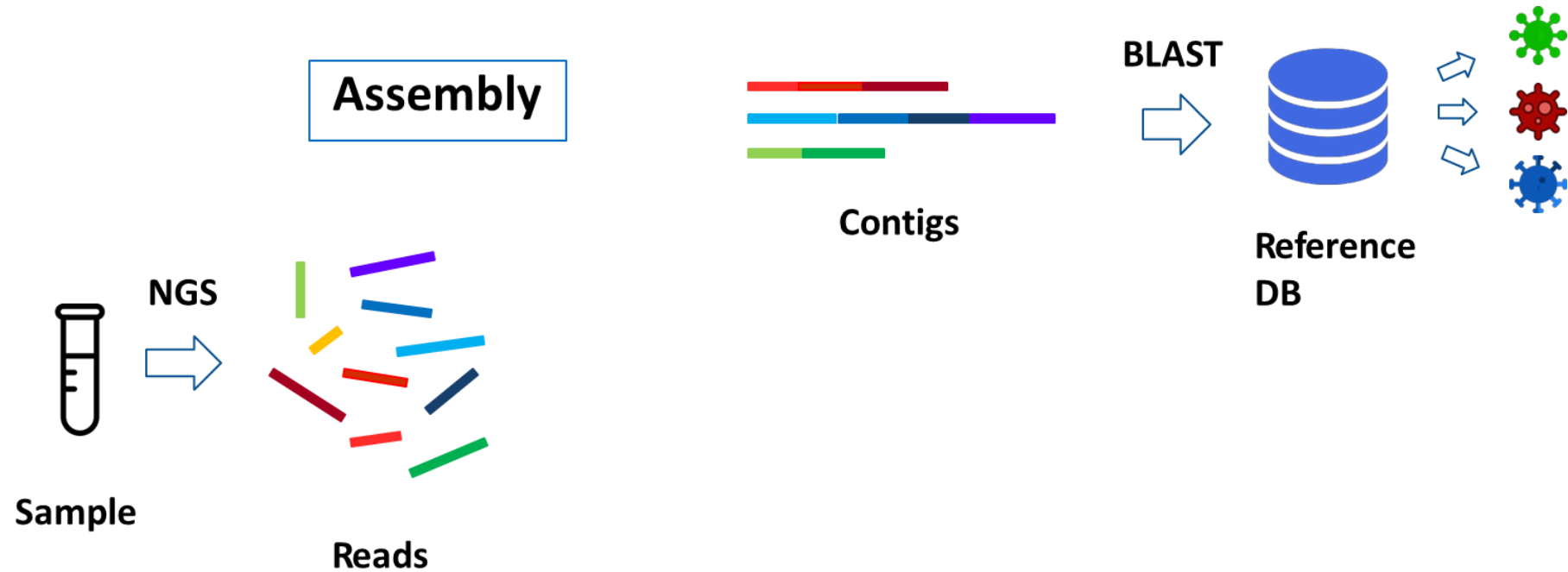


Metagenómica, pipeline de análisis

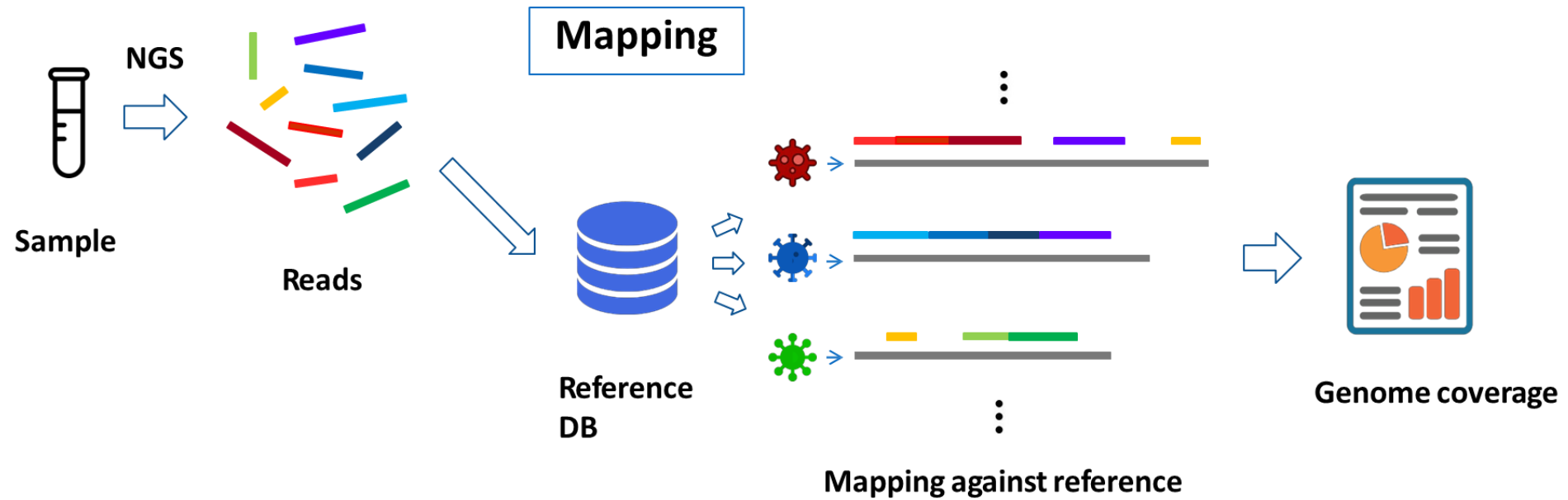


Lysholm et al., Plos One 2012:7,2, e30875

Metagenomic analysis approaches



Metagenomic analysis approaches



Metataxonomics vs Metagenomics (16S vs Shotgun)

Software	Organism	Genetic portion used		Binning algorithm used			Genome coverage	Novel pathogen discovery
		Genetic markers	Whole Genome	Clustering	Mapping	Assembly		
Mothur	Bacteria	X		X			No	No
QIIME	Bacteria	X		X		X	No	No
MEGAN	Bacteria		X			X	No	No
Platypus	Bacteria		X		X		No	No
SURPI	Virus		X			X	No	Yes
Virus-TAP	Virus		X			X	No	Yes
VIP	Virus		X		X		No	Yes
Pathosphere	Virus, Bacteria, Eukarya		X			X	No	Yes

Table 3 Display of features of current bioinformatics pipelines for metagenomic data analysis

Tasks/Pipeline	Quality control	Assembly	Gene detection	Functional annotation	Taxonomic analysis	Comparative analysis	Data management
Clover metagenomics	No	No	Yes	Yes	Yes	Yes	Yes
Galaxy platform ^a	Yes	No	No	No	Yes	Yes	No
IMG/M	No	No	Yes	Yes	Yes	Yes	Yes
MetAMOS	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MG-RAST	No	No	Yes	Yes	Yes	Yes	Yes
RAMMCAP	No	No	Yes	Yes	Yes	Yes	No
Smash Community	Yes	Yes	Yes	Yes	Yes	Yes	No

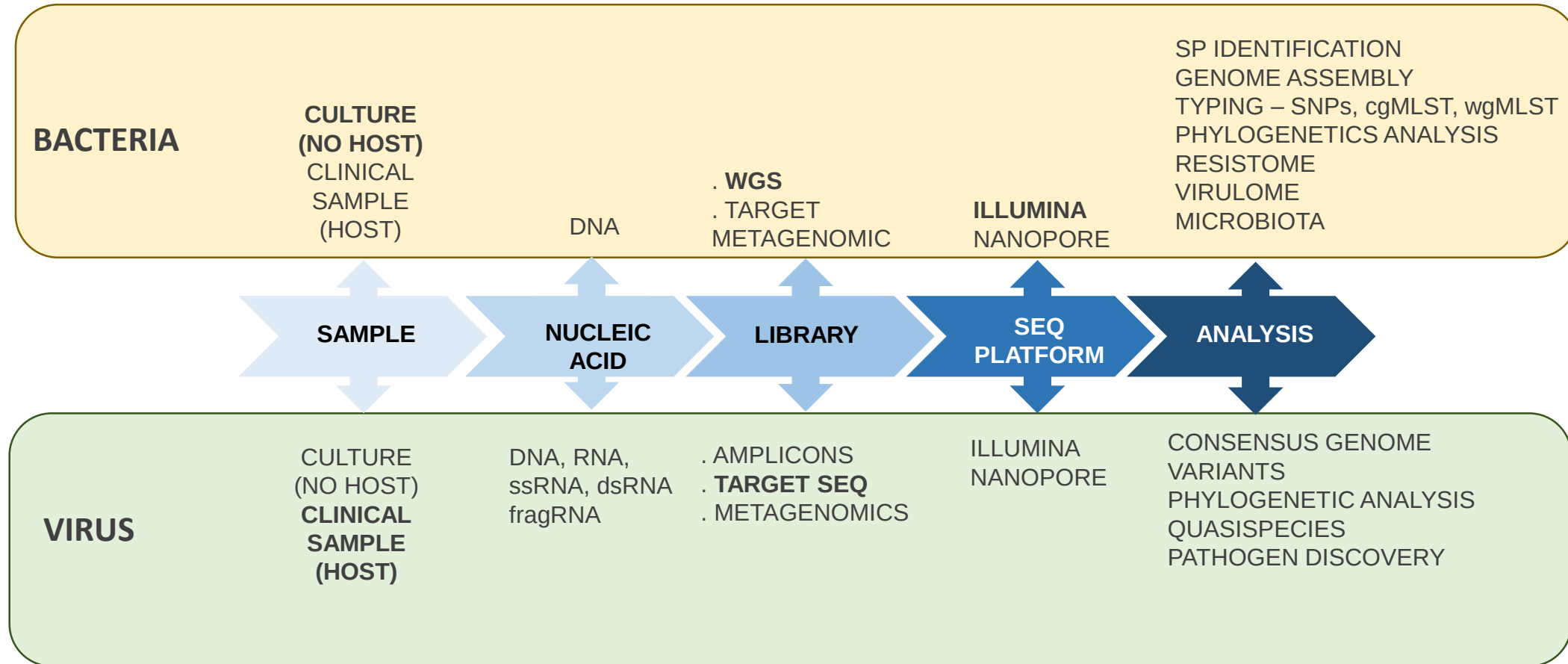
^a Refers to the metagenomics pipeline of galaxy

Navgire et al. Biological Procedures Online 2022

Table 1. Summary and comparison of some commercial kits and pros and cons of different sequencing techniques in COVID-19.

	Shotgun Metagenomics	Capture-Based Enrichment Targeted Sequencing	Amplicon-Based Enrichment Targeted Sequencing
Examples of commercial kits	Illumina Stranded Total RNA Prep with Ribo-Zero Plus	Illumina Respiratory Virus Oligo Panel Illumina Respiratory Pathogen ID/ AMR Enrichment Panel Kit Roche KAPA SARS-CoV-2 Target Enrichment Panel Biosystem TWIST. SARS-CoV-2 Research Panel	Illumina COVIDSeq Test ThermoFisher Ion AmpliSeq™ SARS-CoV-2 Research Panel Paragon Genomics CleanPlex® SARS-CoV-2 Panel Qiagen QIAseq SARS-CoV-2 Primer Panel
Characteristics			
Turnaround time	Long	Moderate	Short
Cost	High	Moderate to Low	Low
The complexity of the workflow	Moderate	Moderate to Low	Low
Coverage of the genome	High	Moderate with high uniformity	Low. with variable uniformity
Sequence depth	Low	High	High
The amount of starting material	High	Moderate to Low	Low
Sensitivity to the target	Low	High	High
Sensitivity to the background	High	Low	Low
Susceptibility to mutational effect	Low	High	High
Applications			
Track transmission	Yes	Yes	Yes
Identification of novel pathogen	Yes	No	No
Identification of co-infections and complex disease	Yes	Only Illumina respiratory panels	No
Identification of new mutations	Yes	Yes	No

Bacterial and Viral Genome Sequencing



Aplicaciones basadas en la preparación de librería

RNA Structure	30
Selective 2'-Hydroxyl Acylation Analyzed by Primer Extension Sequencing (SHAPE-Seq)	31
Parallel Analysis of RNA Structure (PARS-Seq)	32
Fragmentation Sequencing (FRAG-Seq)	33
CXXC Affinity Purification Sequencing (CAP-Seq)	34
Alkaline Phosphatase, Calf Intestine-Tobacco Acid Pyrophosphatase Sequencing (CIP-TAP)	36
Inosine Chemical Erasing Sequencing (ICE)	38
m6A-Specific Methylated RNA Immunoprecipitation Sequencing (MeRIP-Seq)	39
Low-Level RNA Detection	40
Digital RNA Sequencing	42
Whole-Transcript Amplification for Single Cells (Quartz-Seq)	43
Designed Primer-Based RNA Sequencing (DP-Seq)	44
Switch Mechanism at the 5' End of RNA Templates (Smart-Seq)	45
Switch Mechanism at the 5' End of RNA Templates Version 2 (Smart-Seq2)	47
Unique Molecular Identifiers (UMI)	49
Cell Expression by Linear Amplification Sequencing (CEL-Seq)	51
Single-Cell Tagged Reverse Transcription Sequencing (STRT-Seq)	52

Sequencing Methods Review

A review of publications featuring Illumina® Technology

Aplicaciones basadas en la preparación de librería

RNA Transcription	5
Chromatin Isolation by RNA Purification (ChIRP-Seq)	7
Global Run-on Sequencing (GRO-Seq)	9
Ribosome Profiling Sequencing (Ribo-Seq)/ARTseq™	12
RNA Immunoprecipitation Sequencing (RIP-Seq)	15
High-Throughput Sequencing of CLIP cDNA library (HITS-CLIP) or	17
Crosslinking and Immunoprecipitation Sequencing (CLIP-Seq)	17
Photoactivatable Ribonucleoside–Enhanced Crosslinking and Immunoprecipitation (PAR-CLIP)	19
Individual Nucleotide Resolution CLIP (iCLIP)	22
Native Elongating Transcript Sequencing (NET-Seq)	24
Targeted Purification of Polysomal mRNA (TRAP-Seq)	25
Crosslinking, Ligation, and Sequencing of Hybrids (CLASH-Seq)	26
Parallel Analysis of RNA Ends Sequencing (PARE-Seq) or	27
Genome-Wide Mapping of Uncapped Transcripts (GMUCT)	27
Transcript Isoform Sequencing (TIF-Seq) or	29
Paired-End Analysis of TSSs (PEAT)	29

Sequencing Methods Review

A review of publications featuring Illumina® Technology

Aplicaciones basadas en la preparación de librería

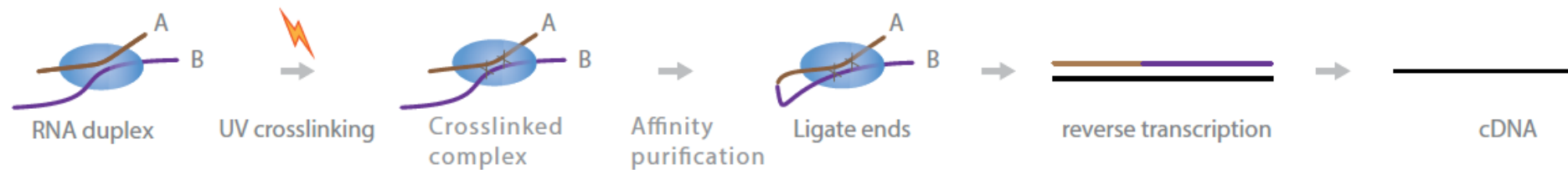
DNA Methylation	63
Bisulfite Sequencing (BS-Seq)	65
Post-Bisulfite Adapter Tagging (PBAT)	70
Tagmentation-Based Whole Genome Bisulfite Sequencing (T-WGBS)	72
Oxidative Bisulfite Sequencing (oxBS-Seq)	73
Tet-Assisted Bisulfite Sequencing (TAB-Seq)	74
Methylated DNA Immunoprecipitation Sequencing (MeDIP-Seq)	76
Methylation-Capture (MethylCap) Sequencing or	79
Methyl-Binding-Domain-Capture (MBDCap) Sequencing	79
Reduced-Representation Bisulfite Sequencing (RRBS-Seq)	81
DNA-Protein Interactions	83
DNase I Hypersensitive Sites Sequencing (DNase-Seq)	85
MNase-Assisted Isolation of Nucleosomes Sequencing (MAINE-Seq)	88
Chromatin Immunoprecipitation Sequencing (ChIP-Seq)	91
Formaldehyde-Assisted Isolation of Regulatory Elements (FAIRE-Seq)	94
Assay for Transposase-Accessible Chromatin Sequencing (ATAC-Seq)	96
Chromatin Interaction Analysis by Paired-End Tag Sequencing (ChIA-PET)	97
Chromatin Conformation Capture (Hi-C/3C-Seq)	99
Circular Chromatin Conformation Capture (4-C or 4C-Seq)	101
Chromatin Conformation Capture Carbon Copy (5-C)	104

Sequencing Methods Review

A BUVISCH publication featuring Illumina® Technology

CROSSLINKING, LIGATION, AND SEQUENCING OF HYBRIDS (CLASH-SEQ)

Crosslinking, ligation, and sequencing of hybrids (CLASH-Seq) maps RNA-RNA interactions¹⁸. In this method RNA-protein complexes are UV crosslinked and affinity-purified. RNA-RNA hybrids are then ligated, isolated, and reverse-transcribed to cDNA. Deep sequencing of the cDNA provides high-resolution chimeric reads of RNA-RNA interactions.



Pros	Cons
• Maps RNA-RNA interactions • Performed in vivo	• Hybrid ligation may be difficult between short RNA fragments

CHROMATIN ISOLATION BY RNA PURIFICATION (CHIRP-SEQ)

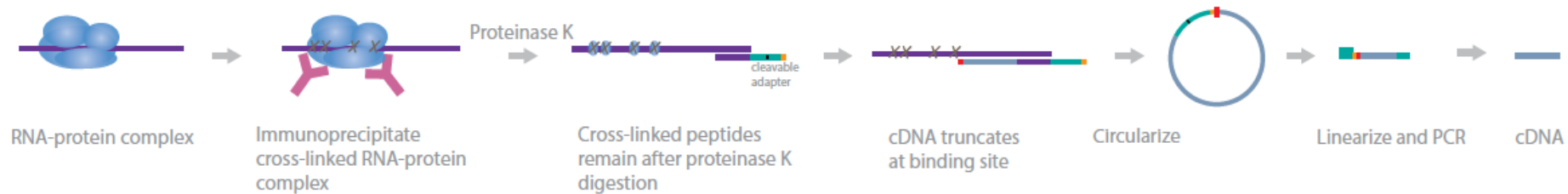
Chromatin isolation by RNA purification (ChIRP-Seq) is a protocol to detect the locations on the genome where non-coding RNAs (ncRNAs), such as long non-coding RNAs (lncRNAs), and their proteins are bound⁷. In this method, samples are first crosslinked and sonicated. Biotinylated tiling oligos are hybridized to the RNAs of interest, and the complexes are captured with streptavidin magnetic beads. After treatment with RNase H the DNA is extracted and sequenced. With deep sequencing the lncRNA/protein interaction site can be determined at single-base resolution.



Pros	Cons
• Binding sites can be found anywhere on the genome • New binding sites can be discovered • Specific RNAs of interest can be selected	• Nonspecific oligo interactions can lead to misinterpretation of binding sites • Chromatin can be disrupted during the preparation stage • The sequence of the RNA of interest must be known

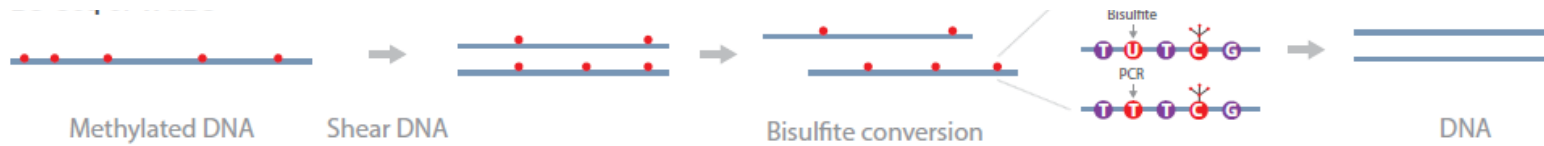
INDIVIDUAL NUCLEOTIDE RESOLUTION CLIP (ICLIP)

Individual nucleotide resolution CLIP (iCLIP) maps protein-RNA interactions similar to HITS-CLIP and PAR-CLIP¹⁵. This approach includes additional steps to digest the proteins after crosslinking and to map the crosslink sites with reverse transcriptase. In this method specific crosslinked RNA-protein complexes are immunoprecipitated. The complexes are then treated with proteinase K, as the protein crosslinked at the binding site remains undigested. Upon reverse transcription, cDNA truncates at the binding site and is circularized. These circularized fragments are then linearized and PCR-amplified. Deep sequencing of these amplified fragments provides nucleotide resolution of protein-binding site.

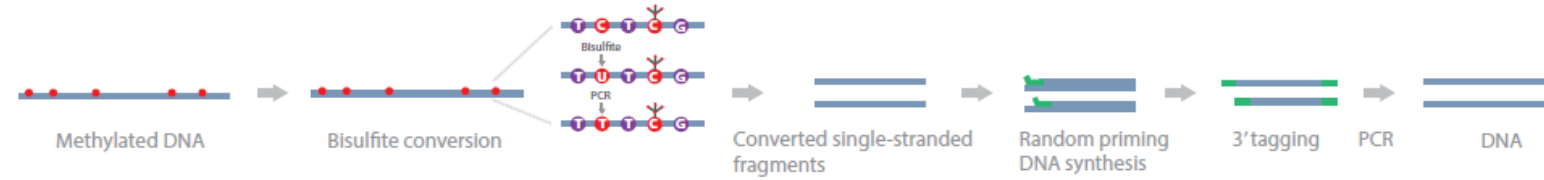


Pros	Cons
• Nucleotide resolution of protein-binding site • Avoids the use of nucleases • Amplification allows the detection of rare events	• Antibodies not specific to target will precipitate nonspecific complexes • Non-linear PCR amplification can lead to biases affecting reproducibility • Artifacts may be introduced in the circularization step

BISULFITE SEQUENCING (BS-SEQ)



EpiGnome Methyl-Seq



Pros	Cons
BS-Seq or WGBS - CpG and non-CpG methylation throughout the genome is covered at single-base resolution 5mC in dense, less dense, and repeat regions are covered	- Bisulfite converts unmethylated cytosines to thymidines, reducing sequence complexity, which can make it difficult to create alignments - NPs where a cytosine is converted to thymidine will be missed upon bisulfite conversion - Bisulfite conversion does not distinguish between 5mC and 5hmC

EpiGnome - Pre-library bisulfite conversion - Low input gDNA (50 ng) - Uniform CpG, CHG, and CHH coverage - No fragmentation and no methylated adapters - Retention of sample diversity	- Bisulfite converts unmethylated cytosines to thymidines, reducing sequence complexity, which can make it difficult to create alignments - SNPs where a cytosine is converted to thymidine will be missed upon bisulfite conversion - Bisulfite conversion does not distinguish between 5mC and 5hmC
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BU-ISCN
Library duplicate percentage

RIBOSOME PROFILING SEQUENCING (RIBO-SEQ)/ARTSEQ™

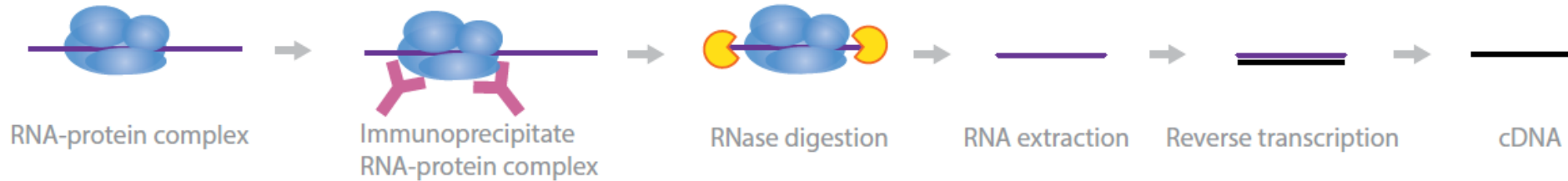
Active mRNA Translation Sequencing (ARTseq), also called ribosome profiling (Ribo-Seq), isolates RNA that is being processed by the ribosome in order to monitor the translation process¹⁰. In this method ribosome-bound RNA first undergoes digestion. The RNA is then extracted and the rRNA is depleted. Extracted RNA is reverse-transcribed to cDNA. Deep sequencing of the cDNA provides the sequences of RNAs bound by ribosomes during translation. This method has been refined to improve the quality and quantitative nature of the results. Careful attention should be paid to: (1) generation of cell extracts in which ribosomes have been faithfully halted along the mRNA they are translating in vivo; (2) nuclease digestion of RNAs that are not protected by the ribosome followed by recovery of the ribosome-protected mRNA fragments; (3) quantitative conversion of the protected RNA fragments into a DNA library that can be analyzed by deep sequencing¹¹. The addition of harringtonine (an alkaloid that inhibits protein biosynthesis) causes ribosomes to accumulate precisely at initiation codons and assists in their detection.



Pros	Cons
• Reveals a snapshot with the precise location of ribosomes on the RNA • Ribosome profiling more closely reflects the rate of protein synthesis than mRNA levels • No prior knowledge of the RNA or ORFs is required • The whole genome is surveyed • Can be used to identify protein-coding regions	• Initiation from multiple sites within a single transcript makes it challenging to define all ORFs • Does not provide the kinetics of translational elongation

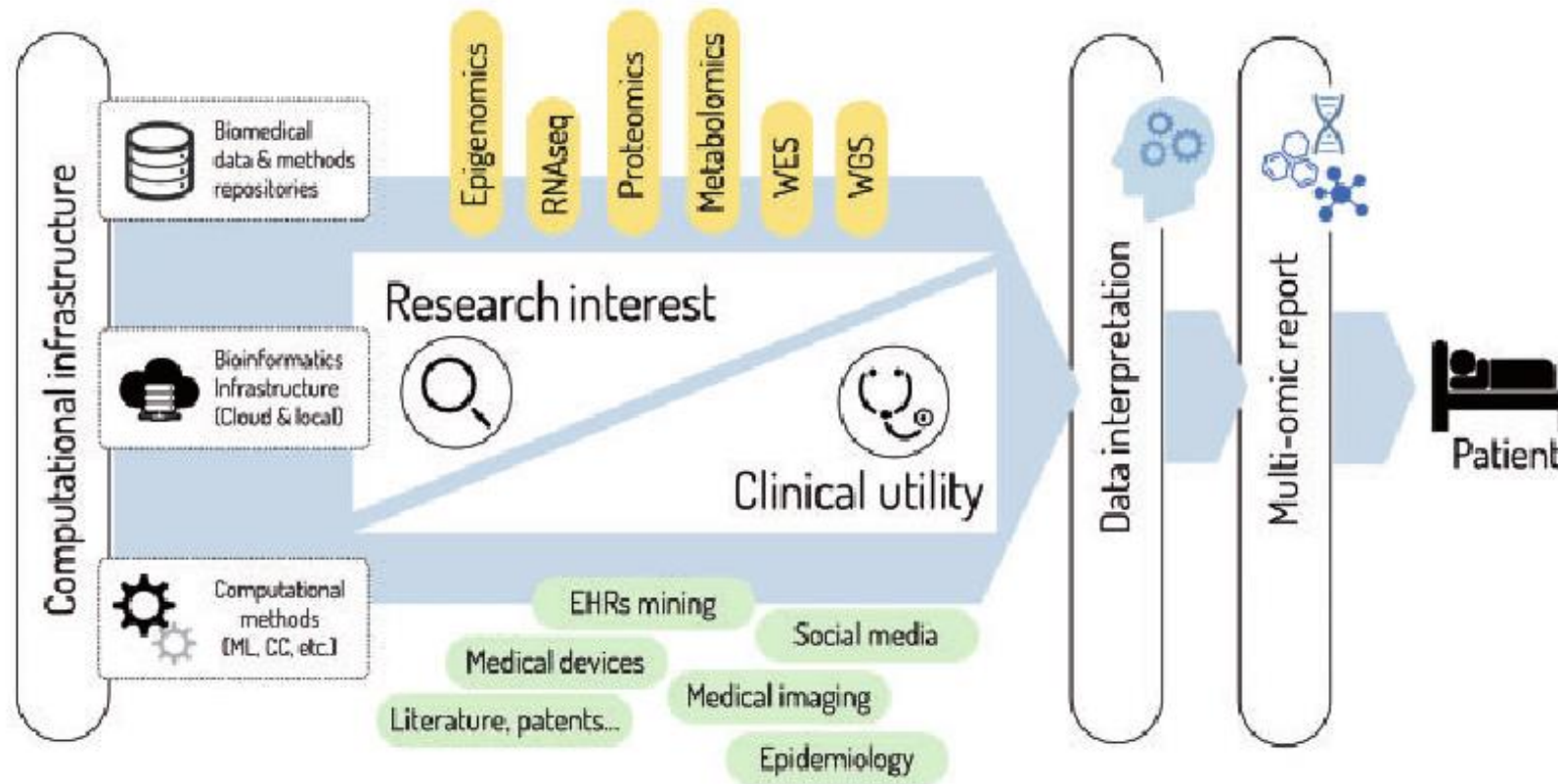
RNA IMMUNOPRECIPITATION SEQUENCING (RIP-SEQ)

RNA immunoprecipitation sequencing (RIP-Seq) maps the sites where proteins are bound to the RNA within RNA-protein complexes¹². In this method, RNA-protein complexes are immunoprecipitated with antibodies targeted to the protein of interest. After RNase digestion, RNA covered by protein is extracted and reverse-transcribed to cDNA. The locations can then be mapped back to the genome. Deep sequencing of cDNA provides single-base resolution of bound RNA.



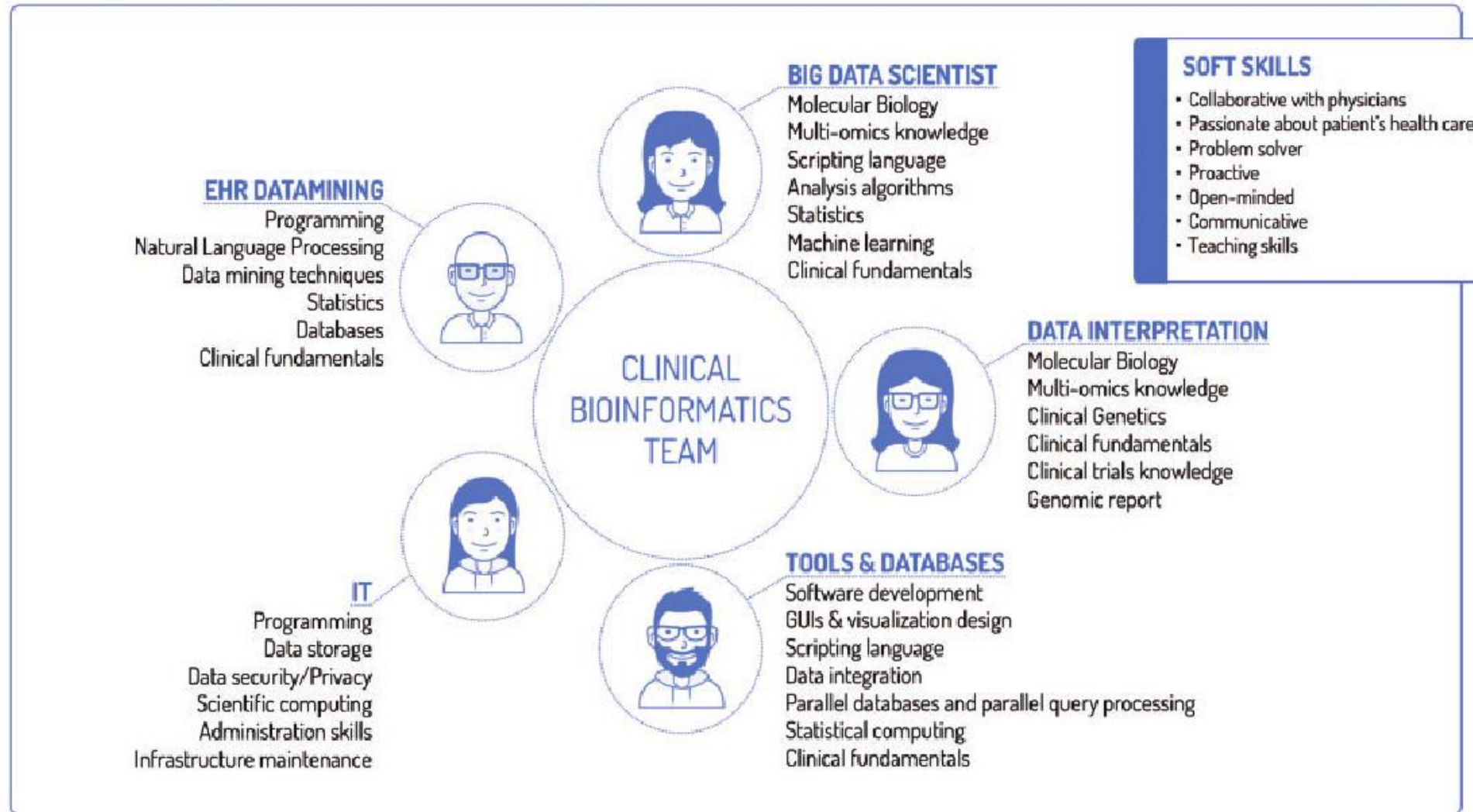
Pros	Cons
• Maps specific protein-RNA complexes, such as polycomb-associated RNAs • Low background and higher resolution of binding site due to RNase digestion • No prior knowledge of the RNA is required • Genome-wide RNA screen	• Requires antibodies to the targeted proteins • Nonspecific antibodies will precipitate nonspecific complexes • Lack of crosslinking or stabilization of the complexes may lead to false negatives • RNase digestion must be carefully controlled

Precision Medicine workflow: from data to patient care



Gómez-López et al., Brief Bioinf 2017, 1-15

Clinical bioinformatics laboratory profile



Skills for Clinical Bioinformaticians

Box 1. Fundamental technical skills for clinical bioinformaticians

1. Informatics

- Experience in UNIX command line.
- A basic programming language (i.e. Python). R as a useful language for handling statistics.
- Knowledge of big data environments.

2. Life sciences

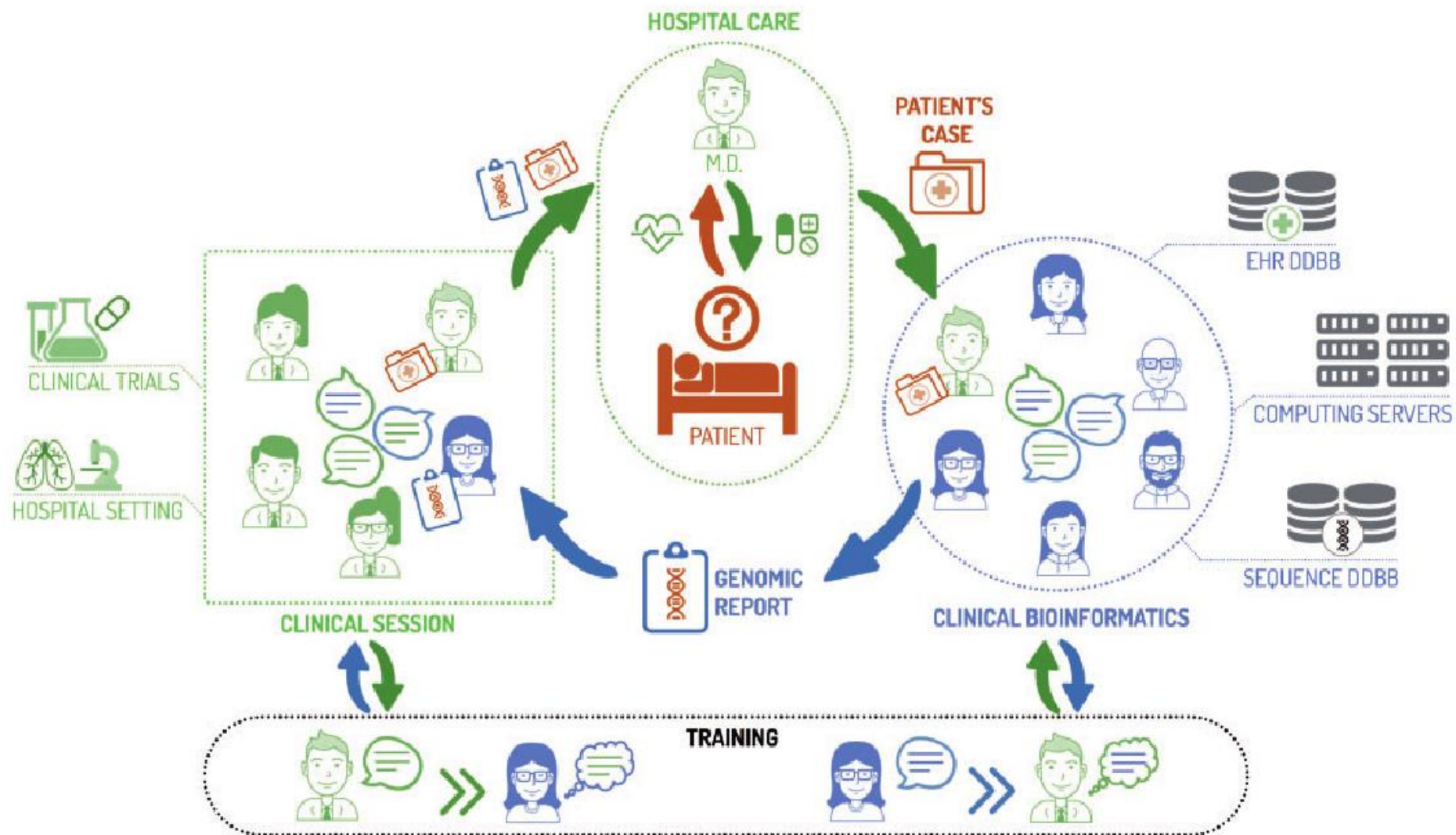
- Understand the different types of biological data and databases.
- Comprehend HTP data analysis methods.
- Multi-omics data integration and interpretation.

3. Clinical scenario

- Be familiar with EHRs, clinical terminology and medical procedures and protocols.
- Get to know medical genomics: diagnosis, predictive and prognosis biomarkers.
- Understand clinical trial design and monitoring.

Gómez-López et al., Brief Bioinf 2017, 1-15

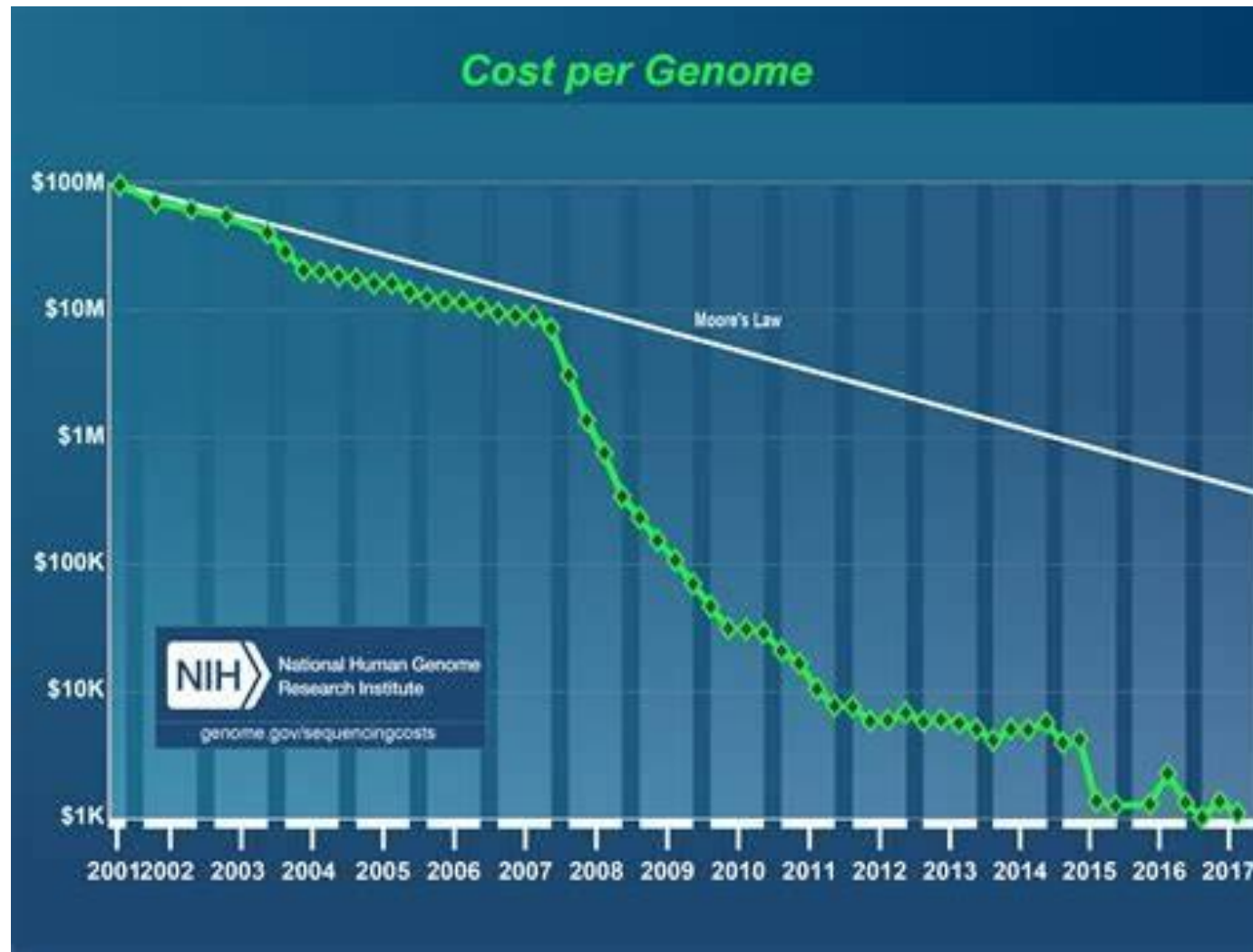
Precision Medicine Workflow in Hospitals



Retos de la Bioinformática en NGS

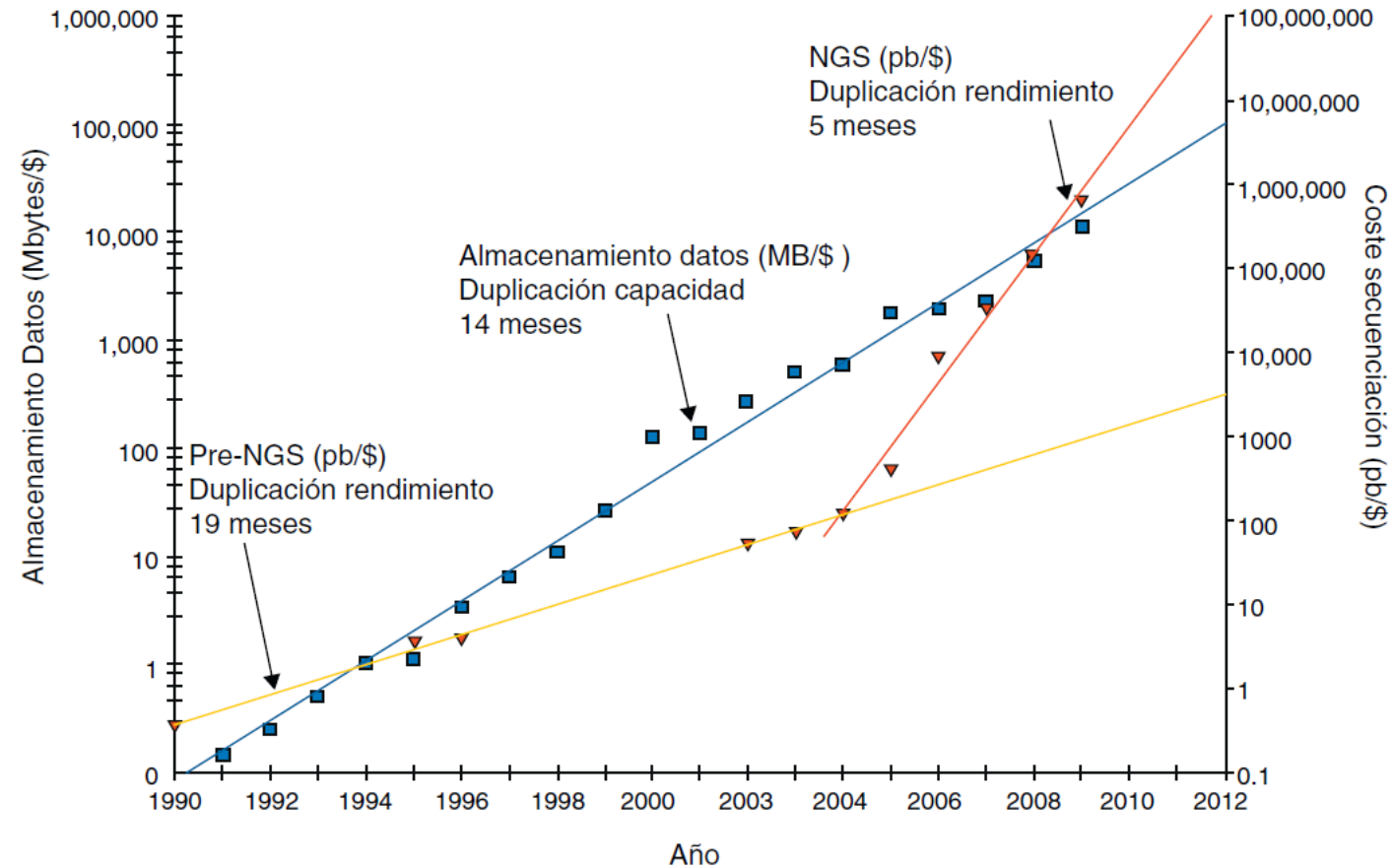
- Tecnología que evoluciona muy rápido
nuevos formatos de ficheros
nuevas aplicaciones
nuevos análisis
- Coste de la secuenciación disminuye
el embudo es el análisis de datos
- Adquisición de secuenciador debe ir ligado
a la compra de computo y contratación
de bioinformático

Coste actual de la secuenciación



BU-ISCI

Costes del almacenamiento vs secuenciación



Adaptada de Stein, Genome Biology 2010, 11:207
BU-ISCI

Retos de la Bioinformática en NGS

- Necesidades de computo
ficheros de gran volumen (10Gb)
elevado uso de CPU y/o memoria
software no comercial en SO Unix
- Necesidades son dependientes de proyecto
No es lo mismo secuenciar un genoma 500Gb
que 50 genomas 25Tb
- Si el proyecto es la aplicación en clínica
Las necesidades de almacenamiento aumentan
por número de pacientes y por tiempo

Retos de la Bioinformática en NGS

- Desarrollo de BD curadas (confianza = reference)
- Algoritmos que resuelvan el problema biológico planteado.
- Necesidades de Bioinformáticos
Análisis de los datos

Softwares comerciales en Bioinformática y NGS

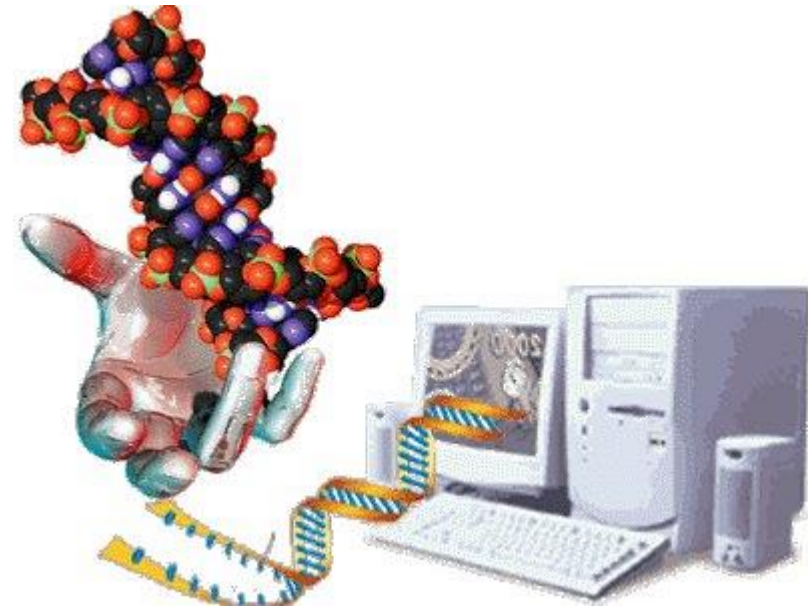
Table I: Examples, features and comparisons of some commonly used commercial bioinformatics software suites

Software	Company	Cost (USD) ^a	Free trial (days)	Platform ^b	NGS analyses ^c	Evolutionary analyses ^d	Database searching ^e	Plug-ins	Workflows	Teaching suitability
Avadis NGS	Strand Scientific Intelligence	\$4500	20	M, W, L	✓	×	×	×	✓	×
CLC Genomics Workbench	CIC bio, Qiagen	\$5500	30	M, W, L	✓	✓	✓	✓	✓	✓
CodonCode Aligner	CodonCode	\$720	30	M, W	✓	✓	×	×	×	✓
Genamics Expression	Genamics	\$295	30	W	×	✓	✓	✓	×	×
Geneious	Biomatters	\$795	14	M, W, L	✓	✓	✓	✓	✓	✓
Full Lasergene Suite	DNASTAR	\$5950	30	M, W	✓	✓	✓	✓	✓	✓
MacVector & Assembler	MacVector	\$300	21	M	✓	✓	✓	×	×	✓
NextGENe	Softgenetics	\$4049	35	W	✓	×	×	×	×	×
Sequencher	Gene Codes	\$2500	30	M, W	✓	✓	✓	✓	×	✓
VectorNTI Advance	Life Technologies	\$600	30	W	×	✓	✓	×	✓	✓

Softwares en Bioinformática y NGS

- Tecnología que evoluciona muy rápido
nuevos formatos de ficheros
nuevas aplicaciones
nuevos análisis
nuevos algoritmos
- Software en continuo desarrollo (Unix)

Gracias por la atención Preguntas ???



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